

OVERALL PROOF ARCHITECTURE FOR THE NAVIER-STOKES STATE-SPACE STRATIFICATION

1. EXECUTIVE SUMMARY

This document explains how the three papers in this folder fit together as a single proof architecture for routing and excluding finite-time singularity scenarios for the three-dimensional incompressible Navier–Stokes equations. The organizing principle is not to guess the shape of a singularity. The organizing principle is to force every possible singular branch into an explicit state-space stratum, then eliminate that stratum by the tool that is strongest for exactly that geometry.

The strategy starts from a finite-energy suitable weak solution

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \quad \nabla \cdot u = 0,$$

and assumes that a terminal singular point $z_* = (x_*, T)$ exists. Local epsilon-regularity gives the first dichotomy:

```
terminal point
|
+-- no scale-critical concentration --> regular by CKN epsilon-regularity
|
+-- positive scale-critical concentration
    |
    +-- local pointwise Type I branch
    |
    +-- local Type II branch
```

The three papers then close the two positive-concentration branches.

Paper	Role in the architecture	Main output
<code>proof_setup.tex</code>	Establishes local concentration, the Type I/Type II dichotomy, Type I Seregin extraction, and the first non-residual Type I class closures.	A local pointwise Type I singularity produces a nonzero normalized bounded centered ancient Seregin profile. That profile is either in a closed non-residual Liouville class or in the residual class.
<code>paperIV_residual_branch.tex</code>	Refines and closes the Type I residual class left by the setup paper.	The retained normalized residual class is empty after an ordered residual decomposition into axisymmetric, rotational, stationary-hull, affine/parasitic, critical-tail, and generic terminal strata.

Paper	Role in the architecture	Main output
<code>type_II_regularity.tex</code>	Excludes the local Type II branch by closing every state-space exit.	Every positive local Type II concentration sequence enters one of the repaired-gauge, multibubble, rough-core, scale-collapse, retained compact-cost, carrier-routing, or terminal-profile alternatives; scale-rigidity, windowwise routing, cost-divergence, analytic-data exhaustion, retained compact-test closure, and critical-profile completeness are all discharged inside the paper.

The resulting architecture is:

```

finite-energy suitable weak solution
|
+-- suppose a singular point exists
|
+-- local CKN concentration gives positive critical mass
|
+-- Type I branch
|   |
|   +-- Seregin ancient limit
|   +-- non-residual Liouville classes closed in proof_setup.tex
|   +-- residual classes closed in paperIV_residual_branch.tex
|   +-- contradiction
|
+-- Type II branch
|   |
|   +-- repaired-gauge/local-profile state-space partition
|   +-- all Type II state-space exits closed in type_II_regularity.tex
|   +-- contradiction

```

This should be read in the hybrid sense used by the papers themselves. The architecture is assertive about the intended proof flow, but it is explicit about the inputs on which each branch depends. The Type I residual closure is now discharged by `paperIV_residual_branch.tex`, including the setup residual hypothesis. In `type_II_regularity.tex`, several items that used to be treated as strategic assumptions have also been converted into theorem or corollary packages: `paper3:prop:local-scale-rigidity-exclusion`, `paper7:cor>windowwise-transition-routing-holds`, `paper7:cor>windowwise-positive-scale-routing-holds`, `paper6a:thm:cost-divergence-exclusion-discharged`, and `paper6a:thm:critical-ns-profile-decomposition-discharged`. The current status is that the Type II exit ledger is closed: entry into the local alternative class, repaired-gauge representation, compactness and local energy propagation, retained compact-test closure, cost/carrier routing, and terminal profile exhaustion are all supplied by named theorem packages in `type_II_regularity.tex`.

2. CURRENT CLOSURE STATUS

All exits in the three-paper state space are now closed.

Branch or interface	Current status	Closing theorem package
Vanishing local concentration	Closed.	<code>p0:thm:ckn</code> , <code>p0:thm:velocity-ckn</code> , <code>p0:thm:no-concentration</code> .

Branch or interface	Current status	Closing theorem package
Local pointwise Type I, non-residual classes	Closed.	p1:thm:classical-classes, p2:thm:NRS, p1:thm:tight-liouville, p1:thm:known-structure-decay-liouville, and the Part III/IV Liouville packages in proof_setup.tex.
Type I residual class	Closed.	thm:paperIV-residual-closure, with cor: setup-residual-hypothesis-proof feeding back into the setup paper.
Type II entry and analytic-data exhaustion	Closed.	paper6:thm:paper0, thm:c1-typeII-branch-exhaustion, paper6:thm:physical-typeII-covered-entry, paper6a:thm:typeII-analytic-data-exhaustive.
Repaired-gauge representation and pressure/local-energy compatibility	Closed.	paper6:thm:representation, thm:c2-representation-implication, thm:c2r-minimal-representation, paper6a:thm:ac-repaired-gauge-consequences.
Compact single core, rough core, multibubble, and cascade exits	Closed.	paper6:thm:single-core, paper6:thm:rough-core, paper6:thm:multibubble, paper7:thm:windowed-h1-failure-exclusion, and the same-point cascade theorems.
Scale collapse and scale-rigidity exits	Closed.	paper6:thm:scale-cost, paper6:thm:scale-stratification, paper6:prop:state-elimination, paper3:prop:scale-rigidity-discharged.
Retained compact cost and carrier exits	Closed.	paper7:thm:classwise-retained-compact-coverage, paper7:thm>windowwise-routing-gives-stratified-routing, paper7:def:stratified-moving-carrier-routing, paper6a:thm:cost-divergence-exclusion-discharged.
Terminal hidden critical profile and residual mass exits	Closed.	paper6a:thm:critical-ns-profile-decomposition-discharged, paper6a:thm:retained-compact-tests-closed, paper6a:thm:terminal-critical-local-compactness, paper6a:thm:expanded-terminal-typeII.
Final Type II assembly	Closed.	paper6:thm:state-decomposition, paper6:thm:unconditional-no-typeII, paper6:thm:local-typeII-exclusion.

The final proof is therefore no longer organized around a list of outstanding strategic assumptions. It is organized around a finite routing graph whose terminal nodes are all contradictions: either local regularity contradicts the choice of singular point, a Liouville theorem contradicts retained

Type I mass, the residual state space is empty, or a Type II exit contradicts retained core mass, finite cost, scale-rigidity closure, or terminal profile exhaustion.

3. COMPLETE EXIT AND REROUTING LEDGER

The tables in this section are the referee-facing ledger for the whole proof. Each row records an exit, the routing test that sends a branch there, whether the exit terminates directly or reroutes to another exit, and the closure mechanism. The entries are intentionally more explicit than the diagrams so that a reader can check the proof linearly.

3.1. Global and Setup Exits.

Exit	Entry test	Routing or rerouting mechanism	Closure
Vanishing CKN concentration	The scale-invariant $C + D$ quantity becomes small at the candidate terminal point.	No further blow-up analysis is permitted; the branch exits to local regularity.	<code>p0:thm:ckn</code> and <code>p0:thm:no-concentration</code> give local boundedness, contradicting singularity.
Vanishing velocity concentration	The velocity-only critical quantity C vanishes along the candidate singular scales.	Routed to velocity epsilon-regularity rather than Seregin or Type II extraction.	<code>p0:thm:velocity-ckn</code> closes the branch.
Positive concentration with local pointwise Type I envelope	Positive local critical mass persists and the Type I scale-invariant L^∞ envelope is finite.	Routed into the Type I bridge sequence and Seregin extraction.	<code>p0:thm:local-weak-serrin-typeI</code> , <code>p1:thm:extraction</code> , and the Type I class ledger below.
Positive concentration with no local pointwise Type I envelope	Positive local critical mass persists but every local Type I envelope fails.	Routed into the Type II local state-space assembly.	<code>p0:thm:typeI-typeII-dichotomy</code> , <code>paper6:thm:paper0</code> , <code>paper6:thm:physical-typeII-covered-entry</code> , and the Type II ledger below.
Terminal velocity no-escape failure	A Type I bridge sequence would lose velocity mass into the terminal slice.	Routed back to the local weak-Serrin/no-escape estimates before extraction.	<code>p0:thm:auto-velocity-no-escape</code> and <code>p0:thm:local-weak-serrin-typeI</code> restore admissibility or close the branch.
Setup residual hypothesis	The setup paper reaches the residual Type I class after direct Liouville classes are removed.	Routed to <code>paperIV_residual_branch.tex</code> as the imported residual object.	<code>thm:paperIV-residual-closure</code> and <code>cor:setup-residual-hypothesis-proof</code> close the setup residual hypothesis.

3.2. Direct Type I Exits.

Exit	Entry test	Routing or rerouting mechanism	Closure
Small-amplitude Seregin profile	The normalized centered ancient profile is small in the relevant scale-invariant topology.	No residual routing; perturbative rigidity applies immediately.	<code>p1:thm:classical-classes</code> forces zero, contradicting retained compact L^3 mass.

Exit	Entry test	Routing or rerouting mechanism	Closure
Stationary L^3 Seregin profile	A time-translate or representative is stationary with critical L^3 control.	Routed to stationary self-similar rigidity.	<code>p1:thm: classical-classes</code> and <code>p2:thm:NRS force zero.</code>
Uniformly L^3 -tight Seregin profile	The ancient trajectory keeps its L^3 mass uniformly tight in space.	Routed through physical pullback and endpoint sequence- L^3 construction.	<code>p2:thm: tight-liouville,</code> <code>p2:thm: AB-endpoint,</code> and <code>p1:thm: tight-liouville force zero.</code>
Structured or decay-covered Seregin profile	Axisymmetry, no-swirl, pointwise scale control, weighted-vorticity smallness, or another listed structure is present.	Routed to the matching structure-specific Liouville theorem.	<code>p3:thm: axisymmetric-liouville,</code> <code>p3:thm: typeI-zero-hypotheses,</code> and <code>p1:thm: known-structure-decay-liouville force zero.</code>
Direct Type I class exhaustion failure	A profile avoids all direct classes.	This is not a terminal gap; it is exactly the residual class.	Routed to the residual ledger by <code>p1:thm: remaining-criterion.</code>

3.3. Type I Residual Exits.

Exit	Entry test	Routing or rerouting mechanism	Closure
Imported residual object mismatch	The residual paper would not be working with the same Seregin object produced by setup.	The imported setup package fixes the object, retained mass, pressure gauge, and ancestry.	<code>thm:imported-setup -results</code> and <code>hyp:base -seregin-hypotheses</code> align the interface.
Non-realized descendant	A recentered limit, hull element, tail state, or terminal profile may not descend from the original branch.	Routed through ancestor-realization and heredity before any closure theorem is applied.	<code>thm:ancestor-realization-inheritance</code> and <code>thm: descendant-heredity</code> make descendant closures logically effective.
Axisymmetric bounded-circulation residual	The residual profile is axisymmetric with bounded centered circulation after lower direct classes are removed.	Routed through centered circulation and axis-absorption; if it falls into no-swirl or lower structured behavior, it is sent back to direct Type I closure.	<code>thm:R1-exclusion</code> closes the exit.
Rotational relative equilibrium	The residual profile is stationary modulo rotation and carries local Coriolis transport structure.	Routed through local Coriolis flux and terminal concentration; lower compact profiles are sent to already closed strata.	<code>thm:terminal-stratification</code> and the rotational routing package close the exit.
Degenerate stationary hull	A time-translation hull contains a degenerate stationary-family carrier with retained concentration.	Routed through stationary-family realization, terminal stratification, and no-active-stationary-carrier closure.	<code>thm:R3-no-attainable-degenerate-family,</code> <code>thm:R3S-terminal-theorem,</code> and <code>thm:R3S-no-active-stationary-carrier.</code>

Exit	Entry test	Routing or rerouting mechanism	Closure
Affine or parasitic quotient	The apparent residual behavior is an affine/parasitic normalization mode rather than genuine nonlinear retained activity.	Routed through affine normalization; oscillatory representatives are removed and non-oscillatory representatives drop to lower strata.	<code>thm:affine-normalization-dichotomy</code> and <code>thm:oscillatory-entry-normalization</code> .
Closed non-affine activity failure	The non-affine active set would fail to be closed under the local observer calculus.	Routed through covariant observers and hull compactness before terminal extraction.	<code>thm:covariant-observer-calculus</code> , <code>thm:closed-nonaffine-activity</code> , and <code>thm:escaping-covariant-hull-compactness</code> .
Recurrent tail-core residual	Tail activity recurs without entering a compact terminal profile.	Routed to recurrent tail-core rigidity or to terminal concentration extraction.	<code>thm:recurrent-tail-core-rigidity</code> and <code>thm:terminal-exhaustion-main</code> .
Log-diffuse critical tail	Critical activity escapes through logarithmic annuli.	Routed to hidden-scale compactification and then to critical-tail compactification.	<code>thm:log-diffuse-from-hidden-scale</code> and <code>thm:critical-tail-compactification-extraction</code> .
Young critical-tail defect	Tail mass survives only as a variance or Young-measure defect.	Routed to first bad mesoscopic scale; realized hidden variance is excluded before compactification.	<code>thm:first-bad-mesoscopic-reduction</code> and <code>thm:no-hidden-scale-variance-realized-effects</code> .
Coherent homogeneous critical tail	Tail compactification produces a coherent homogeneous critical state.	Routed through bounded-origin realization; realized homogeneous tails become incompatible with bounded residual origin behavior.	<code>thm:coherent-critical-tail-classification</code> , <code>thm:bounded-origin-realization-coherent-tails</code> , and <code>thm:realized-critical-tail-rigidity</code> .
Coherent log-periodic critical tail	Tail compactification produces a coherent log-periodic state.	Routed to the same coherent-tail classification and realization test.	<code>thm:coherent-critical-tail-classification</code> and <code>thm:realized-critical-tail-rigidity</code> .
Coherent aperiodic critical tail	Tail compactification produces a compact log-translation hull with no period.	Routed to the aperiodic coherent-tail rigidity branch.	<code>thm:coherent-critical-tail-classification</code> and <code>thm:realized-critical-tail-rigidity</code> .
Terminal nonactive or nonconcentrating state	Terminal extraction produces no active compact concentration.	Routed to terminal nonactive discharge; retained mass cannot disappear.	<code>thm:terminal-inactive-exclusion</code> .
Diffuse terminal defect	Terminal mass is diffuse rather than carried by a compact active profile.	Routed through diffuse compactness, recurrent diffuse measures, and the diffuse trichotomy; active outputs re-enter active terminal routing, affine outputs re-enter affine closure, tail outputs re-enter critical-tail closure.	<code>thm:diffuse-defect-compactness</code> , <code>thm:recurrent-diffuse-defect-measure</code> , and <code>thm:diffuse-defect-trichotomy</code> .

Exit	Entry test	Routing or rerouting mechanism	Closure
Mixed compact-diffuse terminal mass	Compact active concentration and diffuse defect coexist without a clean decomposition.	Routed to the mixed compact-diffuse exclusion.	thm:no-mixed-compact-diffuse.
Finite separated active profile family	Terminal mass splits into finitely many separated compact active profiles.	Routed through separated-profile analysis and indecomposability.	thm:no-finite-separated-profile-family.
Infinite active descendant chain or recurrence loop	Retained activity keeps regenerating through descendants without terminal closure.	Routed to active path-space recurrence and compact active descendant extraction.	thm:active-path-space-recurrence and thm:compact-active-descendant.
Generic terminal residual	No lower residual, tail, affine, diffuse, separated, or recurrent exit remains.	Routed through parasitic-free mildness, sequence- L^3 completeness, and endpoint Liouville.	thm:mildness-inheritance-main, thm:AB-main, thm:R4-final-closure, and thm:paperIV-residual-closure.

3.4. Type II Exits.

Exit	Entry test	Routing or rerouting mechanism	Closure
Type II analytic-data failure	A positive Type II branch lacks the analytic data needed for the local assembly.	Routed through the C1/C3/C4/C5 PDE-form classification until either analytic data are produced or a cost/annulus alternative fires.	thm:c1-typeII-branch-exhaustion, cor:c1-ordered-exhaustion, paper6a:thm:typeII-analytic-data-exhaustive.
Repaired-gauge representation failure	A retained core cannot be put into a stable scale-center chart.	Routed to minimal representation, multibubble, cascade, or scale-rigid alternatives.	paper6:thm:representation, thm:c2-representation-implication, thm:c2r-minimal-representation.
Pressure or local-energy incompatibility in repaired variables	The transformed branch might fail to remain suitable or pressure-stable.	Routed through pressure decomposition, AC repaired-gauge consequences, and local Caccioppoli estimates.	paper2:thm:pressure-decomp, paper2:thm:A, paper2:thm:B, paper2:thm:C, paper6a:thm:ac-repaired-gauge-consequences.
Compact single-core branch	One retained core persists with compact-window control and vanishing localized dissipation.	If the limit is zero, retained mass is lost; if nonzero bounded, the branch exits Type II into Type I-type bounded behavior.	paper1:thm:single-core-criterion and paper6:thm:single-core.
Windowed H^1 or rough-core failure	A selected core has critical mass but lacks inner gradient control.	Routed through Caccioppoli; failure of outer compact-cylinder control re-enters compactness failure, multibubble, or cascade exits.	paper4:thm:caccioppoli, paper7:thm>windowed-h1-failure-exclusion, paper6:thm:rough-core.

Exit	Entry test	Routing or rerouting mechanism	Closure
Same-point comparable-scale multibubble	Several active profiles occur near the same point and comparable scales.	Routed through active-frame reduction, thresholded mass counting, and terminal decoupling.	<code>paper3:thm:multi,</code> <code>paper8:thm:multibubble-frame-reduction,</code> <code>paper6:thm:multibubble.</code>
Same-point strict scale cascade	Active profiles occur at the same point on separated scales.	Routed through same-point reduction, scale separation, and terminal decoupling; irregular cascades reroute to rough-core or scale-rigid exits.	<code>paper3:thm:same-point-reduction,</code> <code>paper8:thm:same-point-scale-separation,</code> <code>paper8:thm:terminal-decoupling-implies-local-decoupling.</code>
Separated-point multibubble	Active profiles occur at separated centers.	Routed through local pressure and mass decoupling; each component returns to compact-core, terminal-profile, or small-data closure.	<code>paper3:thm:separated-point-reduction,</code> <code>paper8:thm:terminal-decoupling-implies-local-decoupling.</code>
Gauge-degenerate branch	The selected chart is nonunique or unstable because several cores or scales compete.	Routed to multibubble, same-point cascade, separated-profile, or scale-rigid exits.	Closed by <code>paper6:thm:state-decomposition</code> after the target exit closes.
Finite scale-collapse cost	The selected scale collapses but the signed or absolute scale cost is finite.	No reroute: retained core mass forces infinite cost.	<code>paper5:thm:cost-exclusion,</code> <code>paper6a:thm:scale-collapse-drift-cost,</code> <code>paper6:thm:scale-cost.</code>
Collapse core-floor failure	A collapsing branch would fail to retain a nonzero localized core.	Routed to vanishing critical L^3 or terminal compactness closure.	<code>paper5:thm:collapse-core-floor,</code> <code>paper6a:thm:collapse-core-floor,</code> <code>paper6a:lem:typeII-concentration-extraction-routes.</code>
Thin-drift scale-collapse alternative	No fixed-length selected window carries enough drift for autonomous extraction.	Routed to the scale-collapse first-failure structure and then to compactness, rough-core, or carrier exits.	<code>paper5:thm:scale-collapse-first-failure-closure</code> and <code>paper5:thm:scale-collapse-alternatives.</code>
Autonomous-modulation alternative	Thick windows exist but modulation fails to converge cleanly.	Routed to autonomous reduced-limit extraction or compactness failure.	<code>paper5:thm:autonomous-modulation-selection,</code> <code>paper5:thm:autonomous-limit,</code> <code>paper5:lem:autonomous-scale-collapse-is-scale-rigid.</code>
Nonzero autonomous reduced limit	A nonzero autonomous or scale-rigid terminal state survives.	Routed to scale-rigid single-core compatibility, scale-rigid stratum decomposition, or compact single-core behavior.	<code>paper3:thm:scale-rigid-single-core-compatibility,</code> <code>paper3:thm:scale-rigid-stratum-decomposition,</code> <code>paper3:prop:scale-rigidity-discharged.</code>

Exit	Entry test	Routing or rerouting mechanism	Closure
Scale-rigid stratum decomposition	Scale-collapse reaches a scale-rigid terminal state but its subtype is not yet identified.	Routed into the finite scale-rigid sub-strata and then to virial, compact-core, or multibubble closures.	<code>paper3:thm:scale-rigid-stratum-decomposition</code> and <code>paper3:prop:scale-rigidity-discharged</code> .
Critical tightness failure	Critical mass tries to escape outside every compact window.	Routed through non-tight exclusion and tail windowed control; surviving mass re-enters terminal or exterior routing.	<code>paper7:thm:critical-tightness-criteria</code> .
Retained compact cost channel	The branch passes compact-cost validation and carries a compact cost-divergence conclusion.	Routed to local cost exclusion and then carrier completion.	<code>paper7:thm:local-cost-to-exclusion</code> , <code>paper7:thm:carrier-validation-retained-stratum</code> , <code>paper7:thm:classwise-retained-compact-coverage</code> .
Adjacent compact-cost exit	The branch leaves the retained compact cost stratum through an adjacent carrier state.	Reduced to carrier completion and finite-error selection.	<code>paper7:thm:adjacent-closure-reduced-to-carrier-completion</code> , <code>paper7:thm:carrier-completion-reduced-to-finite-error</code> .
Fixed carrier failure	Fixed-cutoff carrier errors fail to be summable.	Routed through fixed carrier closure or annular finite-error selection.	<code>paper7:thm:fixed-carrier-closure</code> , <code>paper7:thm:finite-error-selection-reduced-to-annular</code> .
Moving carrier failure	Moving-cutoff carrier errors fail to be summable.	Routed to the canonical moving components J_1, J_6, J_7 .	<code>paper7:thm:moving-carrier-closure</code> , <code>paper7:thm:carrier-package-closes-precarrrier</code> .
J_1 transition/exterior carrier	The cutoff-transition or shell-leakage component is nonintegrable.	Routed by normalized carrier-state extraction to transition/exterior leakage, then to exterior regularity or transition closure.	<code>paper7:thm>windowwise-transition-routing-certificate</code> , <code>paper7:thm>windowwise-routing-gives-stratified-routing</code> .
J_6 negative scale-drift carrier	The negative scale-drift component is nonintegrable.	Routed to the negative-drift scale-collapse branch.	<code>paper6a:thm:moving-cutoff-scale-collapse-alternative</code> and <code>paper6a:thm:finite-infinite-scale-negative</code> .
J_7 positive scale-shell work carrier	The positive scale-shell work component is nonintegrable.	Routed to positive scale-work, then to scale-rigidity unless an earlier exterior, recentering, or negative-drift label fires.	<code>paper7:thm:positive-scale-certificate-enters-scale-rigid</code> , <code>paper7:thm>windowwise-positive-scale-routing-certificate</code> .

Exit	Entry test	Routing or rerouting mechanism	Closure
Divergent compatible localized cost	Retained compact cost forces divergence on the canonical windowwise schedule.	Corrected monotonicity rules out fully summable errors; the branch enters J_1, J_6, J_7 .	<code>paper6a:thm:cost-divergence-routing-dichotomy</code> and <code>paper6a:thm:cost-divergence-exclusion-discharged</code> .
Vanishing terminal critical L^3 norm	A terminal branch loses critical L^3 mass.	Routed to nontrivial critical core and vanishing-critical exclusion.	<code>paper6a:lem:typeII-concentration-extraCTION-routes</code> and <code>paper6a:lem:terminal-local-critical-mass-routing</code> .
Terminal adjacent label	Terminal compactness produces a local state adjacent to a carrier or scale branch.	Routed by terminal adjacent-label routing to the already closed adjacent exit.	<code>paper6a:thm:terminal-adjacent-label-routing</code> .
Residual terminal mass	A terminal branch retains mass outside the recorded compact/core states.	Routed to admissible residual mass state and then profile extraction.	<code>paper6a:thm:residual-mass-admissible-state</code> and <code>paper6a:thm:critical-ns-profile-decomposition-discharged</code> .
Hidden terminal critical profile	Critical terminal mass remains after known profiles are extracted.	The profile decomposition extracts another profile; Kato-small remainders cannot carry hidden mass.	<code>paper6a:thm:critical-ns-profile-decomposition-discharged</code> , <code>paper6a:thm:kato-small-data-critical-stability</code> .
Retained compact-test closure failure	A retained compact test survives but is not assigned to an existing Type II state.	Routed through retained compact-test closure and terminal local compactness.	<code>paper6a:thm:retained-compact-tests-closed</code> , <code>paper6a:thm:terminal-critical-local-compactness</code> .
Terminal boundedness alternative	Terminal frames are bounded, unbounded, or sampled in incompatible ways.	Routed through bounded terminal sequences, coordinate-frame construction, and orbit sampling.	<code>paper6a:thm:bounded-terminal-sequences-from-l3</code> , <code>paper6a:thm:terminal-boundedness-alternatives</code> , <code>paper6a:thm:representation-gives-terminal-orbit-sampling</code> .
Exterior terminal concentration	Terminal mass escapes to the exterior region rather than a compact core.	Routed to exterior stratification and exterior regularity/no-concentration.	<code>paper8:thm:exterior-stratification</code> , <code>paper8:thm:exterior-regularity-no-concentration</code> , <code>paper8:thm:exterior-regular-removal</code> .

Exit	Entry test	Routing or rerouting mechanism	Closure
Final unassigned Type II state	A branch has passed through all earlier tests without terminal contradiction.	The state decomposition has no remaining open target; the final assembly closes the branch.	<code>paper6:thm:state-decomposition,</code> <code>paper6a:thm:abstract-final-typeII,</code> <code>paper6a:thm:expanded-terminal-typeII,</code> <code>paper6:thm:unconditional-no-typeII,</code> <code>paper6:thm:local-typeII-exclusion.</code>

4. REFEREE NAVIGATION GUIDE

This document is meant to be a map, not a substitute for checking the proofs. A referee can use it to track three questions through the papers:

- (1) Where does a candidate singular branch enter the state space?
- (2) Which theorem routes it to the next smaller alternative?
- (3) Which theorem closes that alternative, and what retained quantity gives the contradiction?

The recommended reading order is:

Pass	What to read	What to verify
1	<code>proof_setup.tex</code> , Parts I–II.	A singular point has positive critical concentration; the Type I/Type II split is exhaustive; a local pointwise Type I branch gives a nonzero normalized Seregin profile.
2	<code>proof_setup.tex</code> , Parts III–V.	The non-residual Type I classes are correctly closed by perturbative, stationary, tight, and structured Liouville mechanisms.
3	<code>paperIV_residual_branch.tex</code> .	The residual class imported from the setup paper is the same object being refined; every residual descendant preserves ancestry and retained mass; the ordered residual alternatives are exhaustive and closed.
4	<code>type_II_regularity.tex</code> , representation and local alternatives.	Every Type II branch has the analytic data needed for repaired gauges, pressure reconstruction, local energy, compactness, rough-core routing, and multibubble/cascade routing.
5	<code>type_II_regularity.tex</code> , cost/carrier and terminal profile sections.	Scale-rigidity, retained compact cost coverage, windowwise carrier routing, cost-divergence exclusion, and terminal critical-profile decomposition are proved under the hypotheses under which the assembly invokes them.
6	Final assembly statements in all three files.	No branch is closed using a theorem before its hypotheses have been produced by an earlier routing step.

The proof has three interface contracts. These are the main places where a reader should check that no hypothesis is lost between papers.

Interface	Source theorem package	Target use	Audit point
Setup to residual Type I	<code>p1:thm:extraction</code> , <code>p1:thm:remaining-criterion</code> , and the residual hypothesis recorded in <code>proof_setup.tex</code> .	<code>thm:imported-setup-results</code> , <code>hyp:base-seregin-hypotheses</code> , and <code>thm:paperIV-residual-closure</code> in <code>paperIV_residual_branch.tex</code> .	The residual object must remain a normalized bounded centered Seregin profile with retained compact L^3 mass, compatible pressure gauge, and valid terminal ancestry.
Setup to Type II	<code>p0:thm:typeI-typeII-dichotomy</code> and the positive concentration results in <code>proof_setup.tex</code> .	<code>paper6:thm:paper0</code> , <code>thm:c1-typeII-branch-exhaustion</code> , <code>paper6:thm:physical-typeI</code> <code>I-covered-entry</code> , and <code>paper6a:thm:typeII-analytic-data-exhaustive</code> in <code>type_II_regularity.tex</code> .	The Type II branch carries the local concentration, local energy, pressure, and scale-window data required by the local alternative class.
Type II internal assembly	Repaired gauge, multibubble, rough-core, scale-collapse, retained-cost, carrier-routing, and terminal-profile theorem packages inside <code>type_II_regularity.tex</code> .	<code>paper6:thm:state-decomposition</code> , <code>paper6:thm:unconditional-no-typeII</code> , and <code>paper6:thm:local-typeII-exclusion</code> .	Each closure theorem is invoked only after the branch has been routed into its stated hypotheses; former assumptions such as scale-rigidity, cost-divergence exclusion, and critical-profile completeness are theorem packages, not primitive final inputs.

For quick tracking, the proof should be read as a chain of retained obstructions:

Retained obstruction	First produced	Must survive through	Final contradiction
Positive CKN or velocity concentration	CKN/velocity epsilon-regularity in <code>proof_setup.tex</code> .	Type I extraction or Type II window selection.	Vanishing alternatives imply regularity and are impossible at a singular point.
Compact local L^3 mass of a Type I Seregin profile	Seregin extraction in <code>proof_setup.tex</code> .	Non-residual Liouville routing and residual descendant extraction.	Any Liouville conclusion $V = 0$, residual emptiness, or endpoint L^3 closure contradicts retained mass.
Residual ancestry	Imported setup package in <code>paperIV_residual_branch.tex</code> .	Recentred limits, tail compactifications, active descendants, and terminal profiles.	A closed auxiliary descendant is meaningful only because it is realized by the original residual branch.

Retained obstruction	First produced	Must survive through	Final contradiction
Retained Type II core mass	Type II entry and repaired-gauge selection in <code>type_II_regularity.tex</code> .	Single-core compactness, multibubble splitting, rough-core routing, scale-cost estimates, and terminal profile extraction.	Finite cost, vanishing terminal norm, compact zero limits, or hidden-profile exhaustion cannot coexist with retained mass.
Canonical carrier cost	Retained compact-cost validation in <code>type_II_regularity.tex</code> .	Corrected monotonicity and windowwise J_1, J_6, J_7 routing.	Divergent compatible cost cannot remain a free terminal branch; it enters transition/exterior, negative-drift, or positive-scale/scale-rigid closures.

The most useful cross-check is this: every edge in the decision diagrams below should correspond either to a named theorem in the register at the end of this document or to a cited external theorem explicitly imported in one of the three papers. Every terminal node should say what contradiction is obtained and which retained obstruction supplies it.

5. LATEST STRATEGY REFINEMENT

The current strategy is sharper than the original "prove one large missing local theorem" picture. The proof now proceeds by reducing each hard interface to a small, named theorem package.

Former strategic bottleneck	Current closed treatment
Setup residual hypothesis for Type I	Discharged in <code>paperIV_residual_branch.tex</code> by the ordered residual decomposition, terminal local concentration, critical-tail compactification, descendant heredity, and endpoint L^3 closure.
Scale-rigidity exclusion in the Type II scale/multibubble layer	Discharged by <code>paper3:prop:scale-rigidity-discharged</code> ; bounded selected-window limits and single-core dissipation are supplied by the selected-sequence theorem packages before the rigidity theorem is invoked.
Section 7 compact cost coverage and adjacent closure	Reduced to carrier completion and then to the canonical windowwise moving family; classwise retained compact coverage is discharged by <code>paper7:thm:classwise-retained-compact-coverage</code> .
Windowwise transition and positive-scale routing	Discharged by normalized carrier-state extraction and local state-space labels: nonintegrable J_1 becomes a transition/exterior carrier state, while nonintegrable J_7 becomes positive scale-work routed to scale-rigidity unless an earlier exterior, recentering, or negative-drift label fires.
Cost-divergence exclusion	Discharged by corrected-monotonicity arithmetic plus the canonical windowwise J_1, J_6, J_7 routing package in <code>paper6a:thm:cost-divergence-exclusion-discharged</code> .
Critical Navier–Stokes profile decomposition	Reduced to cited L^3 profile decomposition/nonlinear profile stability plus internal mass-decoupling, Kato-smallness, and hidden-mass exhaustion lemmas in <code>paper6a:thm:critical-ns-profile-decomposition-discharged</code> .

The main discipline for checking the closed proof is therefore not to invoke these packages as black boxes outside their hypotheses. Cost-divergence is tied to the canonical windowwise moving schedule; carrier routing is tied to normalized carrier states rather than pointwise lower bounds; and the terminal profile theorem is tied to the L^3_σ critical profile framework and its Kato perturbative stability.

6. STATE-SPACE DECISION TREES

The diagrams below give the state-space stratification as a theorem-labelled decision procedure. They are rendered natively in LaTeX/TikZ. They are intentionally redundant with the tables: the tables say what each layer means, while the diagrams show how a candidate branch moves through the proof.

6.1. Global Singularity Routing.

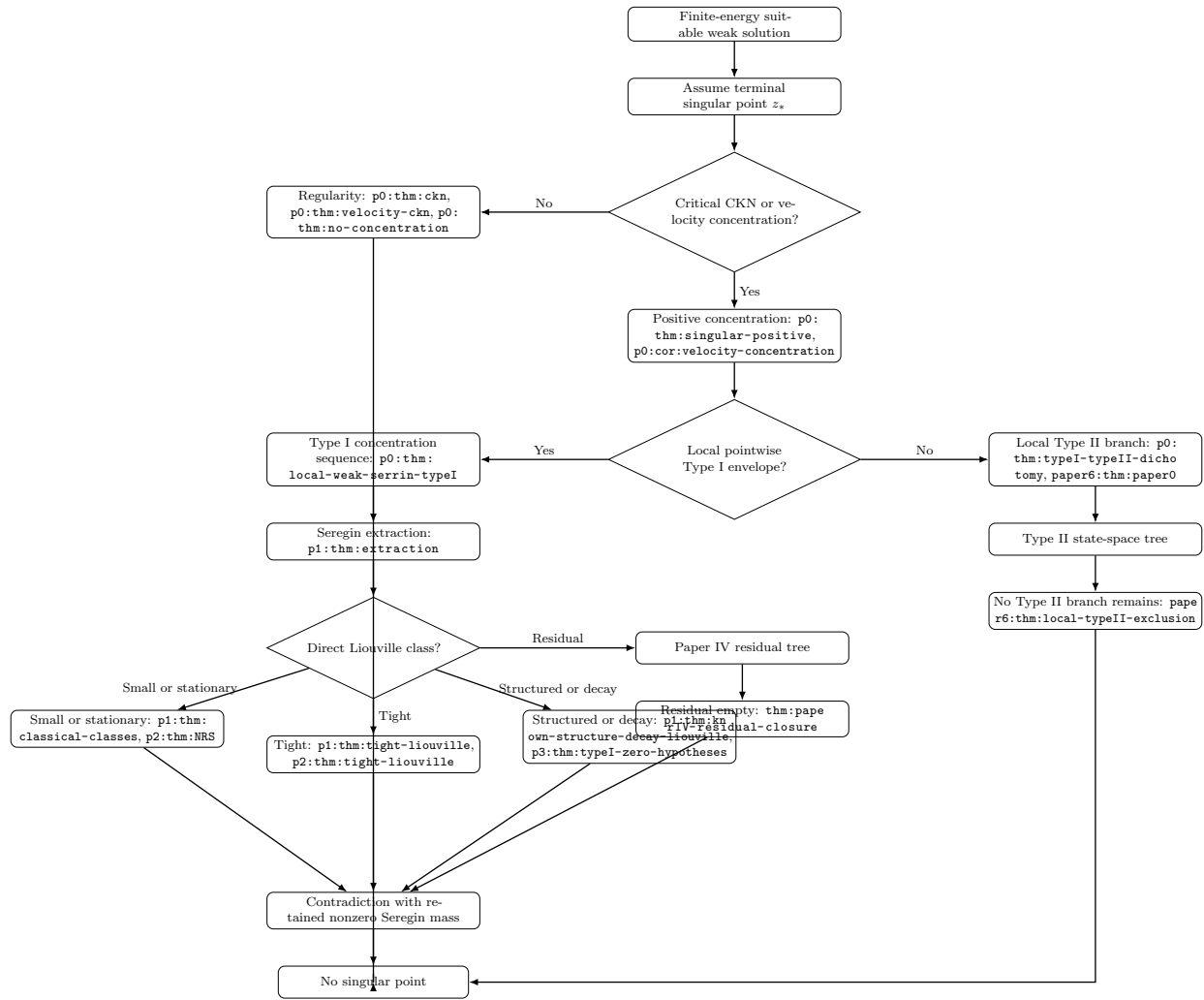


FIGURE 1. Global singularity routing.

6.2. **Type I Residual Decision Tree.**

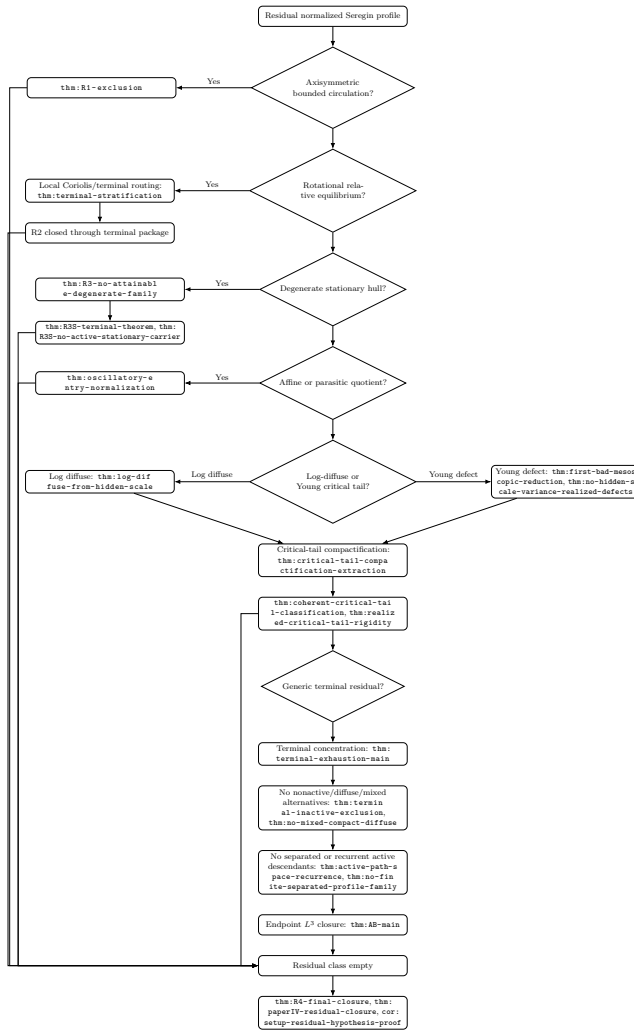


FIGURE 2. Type I residual decision tree.

6.3. Paper IV Residual Branch Detailed Tree. This is the expanded residual branch, matching the internal organization of `paperIV_residual_branch.tex`. The point is that the residual class is not closed by one theorem in isolation: it is routed through lower strata, tail-defect compactification, terminal concentration, descendant heredity, and endpoint L^3 closure.

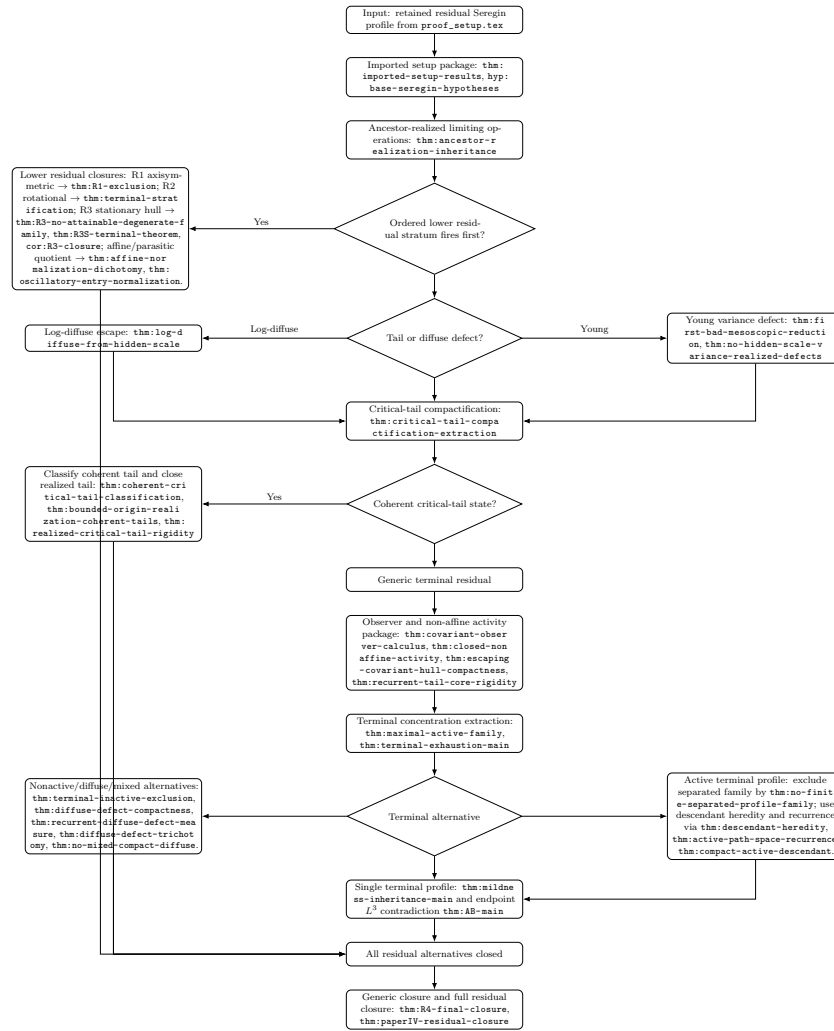


FIGURE 3. Expanded Paper IV residual branch tree.

6.4. **Type II State-Space Decision Tree.**

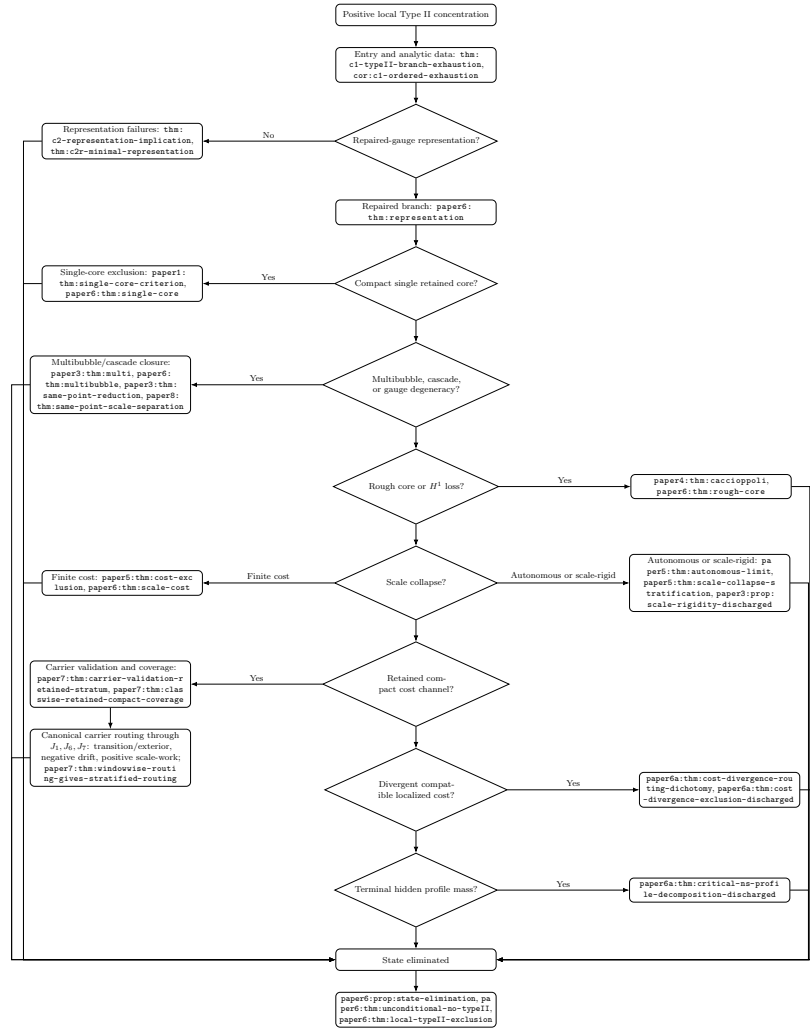


FIGURE 4. Type II state-space decision tree.

6.5. Cost and Carrier Subtree.

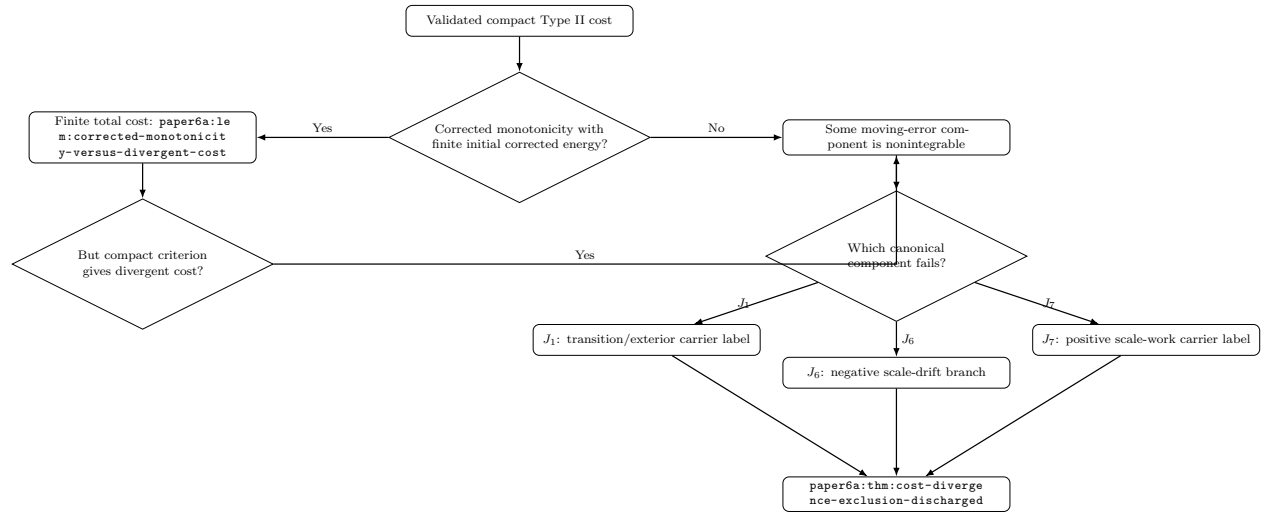


FIGURE 5. Cost and carrier subtree.

7. DIAGRAM EDGE AUDIT

The diagrams above should be read as theorem interfaces, not just pictures. Every edge has a payload: a property that must be proved before the target node can be used. The tables below list those properties explicitly.

7.1. Global Routing Edges.

Edge	What must be verified	Why the edge is valid
Finite-energy suitable weak solution to terminal singular point	The object is a suitable weak solution on a terminal time interval, with the local energy inequality and pressure normalization available.	This is the counterexample setup for <code>p0:thm:ckn</code> , <code>p0:thm:velocity-ckn</code> , and the local concentration package.
Terminal point to CKN/velocity concentration test	The CKN quantities $C(z, r)$, $D(z, r)$, and the velocity concentration quantity are well-defined on shrinking backward cylinders.	The first gate is scale-critical and invariant under Navier-Stokes rescaling.
Concentration test to regularity	Smallness must hold on a cylinder at the candidate singular point.	<code>p0:thm:ckn</code> , <code>p0:thm:velocity-ckn</code> , and <code>p0:thm:no-concentration</code> then give boundedness, contradicting singularity.
Concentration test to positive concentration	There is a sequence of shrinking cylinders carrying a fixed positive critical amount of velocity-pressure or velocity mass.	<code>p0:thm:singular-positive</code> and <code>p0:cor:velocity-concentration</code> supply the retained mass token used later.
Positive concentration to Type I/Type II dichotomy	One must check whether a local pointwise Type I envelope holds near the terminal point.	<code>p0:thm:typeI-typeII-dichotomy</code> makes the split exhaustive.
Type I branch to Seregin extraction	The Type I envelope must give local boundedness away from T , no-escape of velocity mass, local energy control, and pressure compactness.	These are the inputs to <code>p0:thm:local-weak-serrin-typeI</code> and <code>p1:thm:extraction</code> .
Seregin extraction to direct Liouville class test	The extracted profile must be normalized, bounded, centered, ancient, suitable, and nonzero by retained compact L^3 mass.	The class tests only make sense for the normalized Seregin collection.
Direct Liouville class to contradiction	The relevant class hypothesis must force $V = 0$, while retained compact L^3 mass remains positive.	<code>p1:thm:classical-classes</code> , <code>p1:thm:tight-liouville</code> , and <code>p1:thm:known-structure-decay-liouville</code> close these exits.
Direct class failure to residual tree	The profile must fail the small, stationary, tight, and structured/decay tests without losing normalization or retained mass.	<code>p1:thm:remaining-criterion</code> identifies exactly this complement as the residual class.
Residual tree to residual empty	The imported residual object must preserve setup ancestry, retained mass, pressure gauge, and terminal suitability through all descendant operations.	<code>thm:paperIV-residual-closure</code> then empties the residual class.
Type II branch to Type II state-space tree	The branch must carry positive concentration but no local Type I envelope, and must enter the analytic-data/exhaustion framework.	<code>paper6:thm:paper0</code> , <code>thm:c1-typeII-branch-exhaustion</code> , and <code>paper6:thm:physical-typeII-covered-entry</code> supply the entry.

Edge	What must be verified	Why the edge is valid
Type II state-space tree to no Type II branch	Every local Type II state listed by <code>paper6:thm:state-decomposition</code> must be routed to a closed exit.	<code>paper6:prop:state-elimination</code> , <code>paper6:thm:unconditional-no-typeII</code> , and <code>paper6:thm:local-typeII-exclusion</code> finish the branch.

7.2. Type I Residual Edges.

Edge	What must be verified	Why the edge is valid
Residual profile to imported setup package	The profile is the same normalized Seregin object generated by <code>proof_setup.tex</code> , not a newly introduced auxiliary solution.	<code>thm:imported-setup-results</code> and <code>hyp:base-seregin-hypotheses</code> fix the interface.
Imported setup to ancestor-realized operations	Every recentering, blow-down, hull limit, tail compactification, or terminal extraction must be sequence-realized from the original residual branch.	<code>thm:ancestor-realization-inheritance</code> prevents proving closure for unrelated auxiliary limits.
Ancestor-realized operations to ordered lower stratum	The proof must test lower residual strata in the specified order and show earlier strata are absent before later ones are used.	This prevents overlap from reintroducing already closed behavior under another name.
Axisymmetric residual to centered circulation closure	Axisymmetry, bounded centered angular circulation, pressure compatibility, and retained compact mass must survive the centered variables.	<code>thm:R1-exclusion</code> then forces a lower structured class or zero.
Rotational residual to terminal package	The relative-equilibrium structure must produce the local Coriolis flux identity and a terminal concentration recentering.	<code>thm:terminal-stratification</code> routes the branch to lower closed terminal alternatives.
Degenerate stationary hull to R3S closure	The stationary-family hull element must be a sequence-realized Seregin descendant and must still carry retained concentration.	<code>thm:R3-no-attainable-degenerate-family</code> , <code>thm:R3S-terminal-theorem</code> , and <code>thm:R3S-no-active-stationary-carrier</code> close it.
Affine/parasitic quotient to oscillatory normalization	The apparent activity must be shown to be generated by affine/parasitic normalization rather than genuine nonlinear retained mass.	<code>thm:affine-normalization-dichotomy</code> and <code>thm:oscillatory-entry-normalization</code> remove that quotient.
Tail/diffuse defect to hidden-scale reduction	The branch must exhibit log-annular escape or Young/variance defect after compact active alternatives are absent.	<code>thm:log-diffuse-from-hidden-scale</code> and <code>thm:first-bad-mesoscopic-reduction</code> convert hidden escape into a compactified tail state or lower exit.
Hidden-scale branch to critical-tail compactification	The mesoscopic scales must have compactness, pressure gauges, and retained critical activity after quotienting lower affine modes.	<code>thm:critical-tail-compactification-extraction</code> creates the coherent-tail state space.
Coherent critical tail to tail rigidity	The compactified tail must be homogeneous, log-periodic, or aperiodic in the precise log-scale dynamical sense.	<code>thm:coherent-critical-tail-classification</code> and <code>thm:realized-critical-tail-rigidity</code> close realized coherent tails.

Edge	What must be verified	Why the edge is valid
Generic residual to covariant observer calculus	Lower strata, affine/parasitic quotients, and coherent tails must be absent, while non-affine activity remains closed under observers.	<code>thm:covariant-observer-calculus</code> , <code>thm:closed-nonaffine-activity</code> , and <code>thm:escaping-covariant-hull-compactness</code> make terminal extraction legitimate.
Terminal concentration to terminal alternatives	Maximal active families, terminal measures, and compact/diffuse decomposition must be constructed without losing retained mass.	<code>thm:maximal-active-family</code> and <code>thm:terminal-exhaustion-main</code> make the terminal split exhaustive.
Terminal nonactive, diffuse, or mixed branch to closure	One must verify whether retained mass is absent, diffuse, mixed compact-diffuse, or compact active.	<code>thm:terminal-inactive-exclusion</code> , <code>thm:diffuse-defect-trichotomy</code> , and <code>thm:no-mixed-compact-diffuse</code> close the noncompact alternatives or reroute them.
Active terminal profile to descendant/separated tests	The active profile must be checked for finite separated splitting, descendant chains, recurrence, and indecomposable single-profile structure.	<code>thm:no-finite-separated-profile-family</code> , <code>thm:descendant-heredity</code> , and <code>thm:active-path-space-recurrence</code> remove non-single alternatives.
Single terminal profile to endpoint L^3 closure	Parasitic-free mildness and sequence- L^3 completeness must be manufactured from the previous exclusions.	<code>thm:mildness-inheritance-main</code> and <code>thm:AB-main</code> force zero, contradicting retained mass.

7.3. Type II Edges.

Edge	What must be verified	Why the edge is valid
Positive Type II concentration to analytic data	The branch must retain critical mass, local energy, pressure control, scale-window data, and the failure of every local Type I envelope.	<code>thm:c1-typeII-branch-exhaustion</code> , <code>cor:c1-ordered-exhaustion</code> , and <code>paper6a:thm:typeII-analytic-data-exhaustive</code> produce the analytic entry package.
Analytic data to repaired-gauge representation	A nondegenerate core, scale, center, time parametrization, pressure gauge, and modulation coefficients must be selected.	<code>paper6:thm:representation</code> , <code>thm:c2-representation-implication</code> , and <code>thm:c2r-minimal-representation</code> give the modulated local PDE.
Representation failure to multibubble/cascade/scale-rigid exits Repaired branch to compact single core	If the chart is nonunique or unstable, one must identify the competing cores, scales, or degenerating scale-rigid structure. Compact-window bounds, pressure stability, retained mass, and vanishing localized dissipation must all hold.	Representation failure is not terminal; it is routed by <code>paper6:thm:state-decomposition</code> . <code>paper1:thm:single-core-criterion</code> and <code>paper6:thm:single-core</code> then contradict Type II or retained mass.
Repaired branch to rough-core exit	The branch must fail windowed H^1 or compact-cylinder gradient control while keeping critical mass.	<code>paper4:thm:caccioppoli</code> , <code>paper7:thm>windowed-h1-failure-exclusion</code> , and <code>paper6:thm:rough-core</code> reroute roughness to closed compactness/multibubble exits.
Repaired branch to multibubble/cascade	Multiple active profiles, separated centers, same-point scale separation, or chart competition must be detected.	<code>paper3:thm:multi</code> , <code>paper3:thm:same-point-reduction</code> , <code>paper8:thm:same-point-scale-separation</code> , and terminal decoupling close or reroute the split.

Edge	What must be verified	Why the edge is valid
Repaired branch to scale collapse	The selected scale must genuinely collapse relative to the terminal window and the retained core must persist.	Finite-cost exits are closed by <code>paper5:thm:cost-exclusion</code> and <code>paper6:thm:scale-cost</code> ; nonfinite/autonomous exits continue to scale-rigid analysis.
Scale collapse to autonomous or scale-rigid state	Thick selected windows, modulation control, pressure stability, and bounded selected-window limits must be produced.	<code>paper5:thm:autonomous-limit</code> , <code>paper5:thm:scale-collapse-stratification</code> , and <code>paper3:prop:scale-rigidity-discharged</code> close the scale-rigid path.
Repaired branch to retained compact cost channel	The compact tests, retained local mass floor, localized cost comparison, and carrier validation data must all hold.	<code>paper7:thm:carrier-validation-retained-stratum</code> and <code>paper7:thm:classwise-retained-compact-coverage</code> place the branch in the cost/carrier subtree.
Compact cost channel to carrier completion	Fixed and moving cutoff errors must be decomposed into finite-error identities with no unrecorded cost leakage.	<code>paper7:thm:adjacent-closure-reduced-to-carrier-completion</code> and <code>paper7:thm:carrier-package-closes-precarrier</code> reduce the problem to canonical carrier components.
Carrier completion to J_1, J_6, J_7	The automatic moving terms must be summable; only transition, negative drift, or positive scale-shell work may remain nonintegrable.	<code>paper7:thm>windowwise-routing-gives-stratified-routing</code> makes the component list exhaustive.
J_1 to transition/exterior routing	Nonintegrable transition or shell-leakage must produce a normalized carrier state with exterior/transition label.	<code>paper7:thm>windowwise-transition-routing-certificate</code> routes it to exterior or transition closure.
J_6 to negative drift	Nonintegrable negative scale drift must be tied to the canonical moving cutoff and compatible scale-collapse cost.	<code>paper6a:thm:moving-cutoff-scale-collapse-alternative</code> and <code>paper6a:thm:finite-infinite-scale-negative</code> close the negative-drift exit.
J_7 to positive scale-work	Nonintegrable positive scale-shell work must survive normalized carrier extraction and avoid earlier exterior/recentering/negative-drift labels.	<code>paper7:thm:positive-scale-certificate-enters-scale-rigid</code> and <code>paper7:thm>windowwise-positive-scale-routing-certificate</code> reroute it to scale-rigidity.
Divergent compatible cost to cost-divergence exclusion	Corrected monotonicity, finite initial corrected energy, and canonical windowwise compatibility must be checked.	<code>paper6a:thm:cost-divergence-routing-dichotomy</code> and <code>paper6a:thm:cost-divergence-exclusion-discharged</code> force a closed carrier exit.
Repaired branch to terminal hidden profile mass	Terminal frames must be sampled from the represented orbit and bounded in the critical L^3_σ framework.	<code>paper6a:thm:critical-ns-profile-decomposition-discharged</code> extracts all nonzero terminal profiles and rules out hidden mass.
Terminal profile branch to final Type II closure	Retained compact tests, terminal local compactness, expanded terminal data, and boundedness alternatives must all be discharged.	<code>paper6a:thm:retained-compact-tests-closed</code> , <code>paper6a:thm:expanded-terminal-typeII</code> , and <code>paper6:thm:local-typeII-exclusion</code> close the branch.

7.4. Cost and Carrier Subtree Edges.

Edge	What must be verified	Why the edge is valid
Validated compact cost to corrected monotonicity	The localized cost, corrected energy, moving cutoff, and pressure/error terms must satisfy the fixed or moving monotonicity identity.	<code>paper6a:thm:fixed-corrected-monotonicity</code> and <code>paper6a:thm:moving-corrected-monotonicity</code> control the cost balance.
Corrected monotonicity to finite total cost	All error terms must be summable and the initial corrected energy finite.	Then a divergent compact cost contradicts the monotonicity balance.
Failure of finite total cost to nonintegrable component	The fixed and moving carrier decompositions must show that nonintegrability appears in one of the canonical components.	<code>paper7:thm:fixed-carrier-closure</code> and <code>paper7:thm:moving-carrier-closure</code> rule out unclassified error leakage.
Nonintegrable component to J_1, J_6, J_7	Components $J_2, \dots, J_5$ must be automatic or summable on the canonical moving family.	The remaining component determines transition/exterior, negative drift, or positive scale-work routing.
J_1, J_6, J_7 to cost-divergence closure	The selected component must retain its geometric label under normalized carrier-state extraction.	The corresponding routed exit feeds into <code>paper6a:thm:cost-divergence-exclusion-discharged</code> .

8. WHAT IS NOVEL OR INTERESTING TO A PDE EXPERT

The proof architecture has several features that are not just organizational. They change where the hard PDE estimates are used.

Feature	PDE significance
Local state-space stratification before rigidity	The argument avoids requiring a single all-purpose Liouville theorem for bounded ancient solutions. It first routes a candidate into a stratum where a specific compactness, Liouville, virial, or cost mechanism has the right hypotheses.
Retained concentration as the invariant bookkeeping device	Positive CKN or compact L^3 mass is carried through rescaling, gauge changes, terminal extraction, and descendant formation. Every zero or vanishing conclusion is tested against this retained mass.
Type I residual closure by terminal dynamics	The residual Type I problem is not treated as a black-box complement. It is decomposed by terminal active loci, critical tails, diffuse defects, affine/parasitic quotients, and descendant recurrence until endpoint L^3 Liouville becomes applicable.
Repaired-gauge Type II analysis	The Type II branch is analyzed in moving scale-center coordinates with explicit modulation coefficients. Translation drift, scale drift, pressure gauges, and chart degeneracy become named alternatives instead of uncontrolled compactness errors.
Cost-divergence treatment of scale collapse	Genuine Type II scale collapse is not classified by guessing a limiting profile first. Retained local mass forces divergence of a localized scale cost; corrected monotonicity and the canonical windowwise routing package now convert that divergence into exclusion rather than leaving it as a primitive assumption.
No hidden global critical-norm assumption	The residual paper explicitly avoids assuming a global $L_t^\infty L_x^3$ bound for the original residual profile. Sequence- L^3 control is manufactured only after lower strata and parasitic modes are removed.
No hidden global Coriolis–Leray theorem	Rotational relative equilibria are closed by local Coriolis flux and terminal concentration decomposition, not by invoking an unavailable whole-space Liouville theorem for all bounded rotating profiles.
Failure modes become alternatives	Missing pressure compactness, gauge degeneracy, rough-core loss, tail diffusion, critical-tail Young defects, noncompact terminal escape, and nonintegrable carrier components are not left as proof gaps; each is routed into the ordered partition.

For a PDE reader, the most important structural point is that the proof tries to replace a single impossible classification theorem by many smaller rigidity statements whose hypotheses are produced by the preceding routing step. The interesting work is therefore not only in the terminal contradictions; it is in showing that the routing is exhaustive and that descendants remain tied to the original singularity.

9. STRATEGY-LEVEL PORTABILITY BEYOND THIS PARTICULAR PROOF

The architecture can be evaluated at two levels. At the proof level, every estimate, compactness passage, pressure reconstruction, and Liouville invocation must be correct for the three-dimensional Navier–Stokes equations. At the strategy level, many of the ideas are more general: they describe how to organize a hard PDE regularity problem even if a particular implementation in 3D Navier–Stokes later has to be repaired or replaced.

The portable lesson is not ”copy these strata verbatim.” The portable lesson is ”find the invariant observables, route every possible failure mode, and use only the rigidity theorem justified by the stratum that has actually been reached.”

Strategy-level item	Generality as a PDE technique	What changes in another PDE
Critical concentration as the first gate	Very general for scale-critical regularity problems. Analogues appear whenever an epsilon-regularity or small-data criterion detects regularity below a critical threshold.	Replace CKN $C + D$ by the problem’s critical Morrey, Strichartz, energy, entropy, curvature, vorticity, or defect-measure quantity.
Type I/Type II separation	Broadly useful when there is a natural self-similar blow-up rate. It separates controlled self-similar compactness from genuinely supercritical dynamics.	The Type I norm and rate must be the one adapted to the equation: curvature for geometric flows, critical spacetime norms for dispersive equations, gradient or energy density for heat flows.
Blow-up extraction into ancient or eternal objects	Very general. Many regularity problems reduce singularity formation to ancient solutions, tangent flows, bubbles, or profiles.	The compactness topology changes: suitable weak limits for Navier–Stokes, Brakke/tangent flows for mean curvature, concentration compactness profiles for dispersive PDE, Uhlenbeck limits for gauge theory.
Retained mass bookkeeping	Very general and highly valuable. A contradiction proof needs a nonzero quantity that survives all limits and cannot coexist with a zero/rigid conclusion.	The retained token may be energy, entropy drop, curvature concentration, enstrophy, charge, topological degree, L^2 mass, or a profile norm.
Ordered state-space stratification	Very general. It is a way to replace a missing global classification theorem by many smaller conditional classifications.	The strata must be chosen from the equation’s genuine failure modes: radiation, bubbling, necks, solitons, cascades, boundary concentration, oscillation, defect measures, or symmetry reductions.
Classification versus closure	Very general. Decomposition and exclusion should be separate statements, so an unclosed stratum is visible rather than hidden.	Another PDE may have a complete decomposition but only partial closures; the architecture still identifies exactly which theorem is missing.

Strategy-level item	Generality as a PDE technique	What changes in another PDE
Failure modes as named alternatives	Very general and often underused. A failed compactness estimate, failed gauge, or failed pressure bound should produce a new branch of the proof, not an implicit assumption.	The named alternatives depend on the PDE: gauge bubbling in Yang–Mills, neck regions in harmonic map heat flow, radiation channels in dispersive PDE, shock/oscillation measures in conservation laws.
Repaired gauges and modulation parameters	General for problems with symmetries or moving concentration centers. This is common in soliton dynamics, geometric flows, dispersive blow-up, and gauge theories.	The parameters may be translation, scale, phase, Galilean boost, rotation, gauge transform, reparametrization, or diffeomorphism. The orthogonality conditions and modulation equations are equation-specific.
Pressure/gauge bookkeeping	Specific in form but general in spirit. Any PDE with nonlocal constraints, gauge freedom, or Lagrange multipliers needs compatible normalization across limits.	Replace pressure gauges by Coulomb gauges, DeTurck gauges, harmonic coordinates, elliptic potentials, chemical potentials, or constraint multipliers.
Cost-divergence or monotonicity-defect arguments	Broadly useful when a branch can escape by scale drift or migration. A localized cost can turn "escape through scales" into a quantitative contradiction, provided the error terms are either summable or routed as named carrier states.	The cost may be entropy dissipation, frequency drop, Morawetz flux, monotonicity defect, action, curvature scale drift, virial defect, or modulation energy.
Tail compactification and critical-tail analysis	General for noncompact critical problems where mass can escape to infinity or to logarithmic scales.	The compactification variable changes: physical infinity, frequency infinity, logarithmic scale, null infinity, neck length, or renormalized time.
Active-locus extraction and descendant heredity	Very general for concentration compactness. If a proof extracts secondary profiles, it must prove they are descendants of the original counterexample and still carry the retained obstruction.	The descendant relation may be profile decomposition ancestry, tangent-flow ancestry, bubble-tree ancestry, or gauge-recentered limit ancestry.
Endpoint rigidity after manufacturing its hypotheses	Very general. Instead of assuming a strong global critical bound, one can try to stratify away all failures until an endpoint theorem becomes applicable. The current papers use this both in the Type I residual endpoint L^3 closure and in the Type II critical-profile/Kato package.	The endpoint theorem may be a Liouville theorem, scattering criterion, backward uniqueness theorem, monotonicity equality case, soliton rigidity theorem, or no-neck theorem.
Residual class as a real object, not a wastebasket	Very general. The complement of known classes should be refined until it has its own geometry.	The residual geometry will be problem-specific: diffuse measures, radiation, soliton trains, neck cascades, parasitic gauge modes, or weak turbulence.

This strategy-level viewpoint remains useful even if one later finds that a specific estimate in the 3D Navier–Stokes implementation is wrong. A failed estimate would not invalidate the architectural lesson automatically; it would identify one of three things:

- (1) the proposed stratum is not actually closed by the stated tool;
- (2) the routing into that stratum is not exhaustive;
- (3) the observable used to define the stratum is not stable enough under the

relevant compactness or gauge operation.

Those are useful failure modes. They tell a PDE analyst where the proof must be repaired: strengthen the observable, split the stratum further, prove a missing rigidity theorem, or accept that the branch is a genuine remaining obstruction. In this sense, the architecture is valuable not only as a claimed proof route but also as a diagnostic framework for regularity programs.

10. THE CENTRAL IDEA: STRATIFY THE STATE SPACE BEFORE TRYING TO PROVE RIGIDITY

The main efficiency gain is that the proof does not seek one universal Liouville theorem for every bounded ancient solution or one universal description of every possible blow-up. Instead it uses a decision procedure. At every stage the proof asks a scale-invariant question whose answer places the candidate singular branch into a smaller state space.

The state space is partitioned by features that are stable under the natural Navier–Stokes rescalings:

- (1) Does the candidate singular point carry critical CKN mass?
- (2) Does the branch satisfy a local Type I envelope, or does every local Type I envelope fail?
 - (1) If Type I, what kind of centered ancient Seregin profile is produced?
 - (2) If the Seregin profile is not in a known Liouville class, which residual terminal geometry does it have?
- (1) If Type II, does the concentration have one retained core, several cores, a cascade, a rough core, scale collapse, or a scale-rigid terminal state?
- (1) If the branch escapes compactness, is the escape radiative, diffuse, critical-tail, affine/parasitic, or active and recurrent?

The point of this ordered classification is that each negative result only has to be proved on the stratum where its hypotheses are naturally true. For example, endpoint L^3 Liouville theory is used only after the argument has converted a profile into a sequence- L^3 or uniformly tight state. The axisymmetric circulation equation is used only after the profile has been placed in the axisymmetric bounded-circulation stratum. Repaired gauges are used only for Type II branches where the selected core has moving scale and center. Cost identities are used only after genuine scale collapse has been selected.

The partition is ordered rather than naively disjoint. A profile may satisfy several descriptive properties. The proof assigns it to the first applicable closed stratum. This prevents overlap from creating ambiguity and prevents lower-order or already-closed behavior from being reintroduced later under a different name.

The actual observables used for routing are PDE observables, not formal labels.

Observable	What it detects	Where it is used
$C(z, r)$ and $D(z, r)$	Local critical velocity-pressure concentration.	First concentration gate and Type II entry.
Local pointwise Type I envelope	Whether parabolic rescaling gives bounded ancient compactness.	Type I/Type II dichotomy.
Compact local L^3 nonvanishing	The mass token that prevents Liouville zero conclusions from being harmless.	Type I Seregin limits and terminal residual states.
Uniform L^3 -tightness or sequence- L^3 control	Whether endpoint ancient Liouville theory can be applied.	Tight Type I class and final generic residual closure.
Stationarity, relative equilibrium, axisymmetry, circulation, swirl	Special structures that activate known or local Liouville mechanisms.	Structured Type I classes and lower residual strata.
Scale-center modulation a, b	Whether a Type II core is drifting, collapsing, or approaching an autonomous regime.	Repaired-gauge Type II analysis.

Observable	What it detects	Where it is used
Local windowed H^1 control	Whether compactness can be upgraded from critical control.	Rough-core and compact Type II branches.
Active loci and diffuse defect measures	Whether terminal mass is compact, separated, diffuse, or hidden between scales.	Residual Type I closure.
Localized scale cost	Whether scale collapse can coexist with retained mass.	Type II scale-collapse exclusion.

11. THE FIRST GATE: LOCAL CONCENTRATION AND THE VANISHING BRANCH

The first paper, `proof_setup.tex`, begins with the local theory around a candidate singular point. For $z_0 = (x_0, T)$ and $r > 0$, the basic scale-invariant quantities are

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

and

$$D(z_0, r) = r^{-2} \int_{T-r^2}^T \int_{B_r(x_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

The Caffarelli–Kohn–Nirenberg epsilon-regularity theorem says that if the critical quantity $C + D$ is sufficiently small on a backward parabolic cylinder, then the solution is locally bounded on a smaller cylinder. Thus a true singular point cannot be invisible to these critical quantities.

This gives the first excluded singularity type.

Candidate type	Meaning	Exclusion mechanism
Vanishing CKN branch	At the putative singular point, the scale-invariant velocity-pressure mass tends to zero on all sufficiently small cylinders.	CKN epsilon-regularity gives local boundedness, contradicting singularity.
Vanishing velocity branch	The velocity-only critical quantity $C(z_0, r)$ tends to zero along the scales relevant to the singular point.	The velocity epsilon-regularity criterion again gives local regularity.

After this gate, every remaining singular point has positive critical mass at arbitrarily small scales. That positive mass is the conserved token that the rest of the proof carries through blow-up, compactness, recentering, and terminal-profile extraction. Whenever a branch claims to disappear, diffuse away, or become perturbative, the argument checks whether the retained mass has actually been accounted for. If the mass cannot be accounted for, the branch is not a singular branch.

12. THE SECOND GATE: TYPE I VERSUS TYPE II

Once positive local concentration is fixed, the next partition is by blow-up rate. A terminal point is in the local pointwise Type I branch when, in some neighborhood of the point,

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} < \infty.$$

It is in the local Type II branch when positive concentration is present but no such local Type I scale-invariant bound holds.

This dichotomy is efficient because the two branches have different natural compactness theories.

Branch	Natural rescaling	Main compact object	Main obstruction
Local pointwise Type I	Parabolic rescaling at the singular scale, followed by centered self-similar variables.	A smooth bounded ancient Seregin profile $V(y, \tau)$.	Bounded ancient solutions are too large a class unless stratified.
Local Type II	Selected local windows around retained concentration cores, with repaired moving scale-center gauges.	A represented local branch (V, P, a, b) solving a modulated Navier-Stokes equation on compact windows.	The core may split, cascade, lose compactness, become rough, or collapse in scale.

The proof deliberately does not merge these branches. The Type I argument is an ancient-solution classification problem. The Type II argument is a local concentration-geometry problem. Treating them separately keeps each tool in the regime where it has a clean scale-invariant meaning.

13. THE TYPE I BRANCH: FROM A SINGULAR POINT TO A BOUNDED CENTERED ANCIENT STATE

The Type I pipeline in `proof_setup.tex` is:

```

local pointwise Type I singular point
|
+-- positive local velocity concentration
|
+-- endpoint weak-Serrin / local Type I bounds
|
+-- terminal velocity no-escape
|
+-- admissible Type I bridge sequence
|
+-- Seregin extraction
|
+-- smooth bounded ancient physical profile U
|
+-- centered variables
|
+-- normalized centered ancient profile V with retained compact L^3 mass

```

The centered variables are

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(x, t),$$

and the centered equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla) V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

The crucial output is not merely that a limit exists. The output is a normalized object with positive retained local mass. In schematic terms, there are compact sets and times for which

$$\int_K |V(y, \tau)|^3 dy > 0$$

in the limiting profile. This nonvanishing condition is what contradicts every Liouville conclusion $V \equiv 0$.

The pressure bookkeeping matters here. The source pressure p , the physical ancient pressure Q , and the centered pressure P are all understood modulo functions of time. This gauge freedom is harmless for the equations and the local energy inequality, but it is essential for compactness: pressure limits must be compared in compatible local gauges rather than as absolute functions.

13.1. Why Seregin Extraction Is Used. Seregin extraction is the right compactness mechanism for Type I because the Type I envelope produces uniform local boundedness away from the terminal time after rescaling. It converts a dynamic singularity into an ancient object on the whole time interval $(-\infty, 0)$, then centered variables turn the self-similar scaling into time translation in τ . This makes the problem a classification problem for bounded ancient trajectories, where Liouville and compact-dynamical tools are effective.

13.2. The First Type I State-Space Partition. The setup paper defines a collection $\mathcal{S}$ of normalized Seregin ancient limits generated by admissible Type I extractions. Each element $V \in \mathcal{S}$ is assigned to one of the following classes:

Type I class	Selection criterion	Why it is removable
Small-amplitude class	The bounded ancient profile is sufficiently small in the relevant scale-invariant topology.	Perturbative bounded ancient Liouville theory forces $V = 0$.
Stationary L^3 class	The centered profile has a stationary representative with critical L^3 control.	The Necas–Ruzicka–Sverak stationary self-similar rigidity theorem excludes nonzero profiles.
Uniformly L^3 -tight class	The profile’s L^3 mass remains uniformly tight in space along the ancient time direction.	Physical pullback gives the endpoint sequence- L^3 structure needed for the Albritton–Barker ancient Liouville theorem.
Liouville-covered structure/decay class	The profile has one of the explicitly listed structures or decay properties covered by known Liouville theorems.	Axisymmetric, swirl, weighted-vorticity, or other cited rigidity mechanisms force zero.
Remaining/residual class	The complement of the previous four classes inside $\mathcal{S}$.	Not closed in the setup paper; refined and closed in <code>paperIV_residual_branch.tex</code> .

The first four classes are not intended to be disjoint. The residual class is the complement of their union after those classes have been removed. This is important: residual means “not already removed by the available direct Liouville mechanisms,” not “mysterious singularity with no structure.”

14. THE NON-RESIDUAL TYPE I TOOLS

The setup paper chooses tools according to the amount of structure already visible in the Seregin limit.

Tool	Where it enters	Why it is chosen
Endpoint weak-Serrin local Type I estimate	Before Seregin extraction.	It prevents velocity mass from escaping into the terminal time slice and supplies the local energy control needed for admissibility.
Local compactness and pressure normalization	During blow-up extraction.	It provides a suitable ancient limit and keeps pressure meaningful modulo functions of time.
Centered self-similar variables	After physical ancient extraction.	They convert the Type I scaling symmetry into an autonomous centered ancient equation.
Perturbative Liouville theorem	Small-amplitude class.	Small bounded ancient solutions cannot carry retained concentration.
NRS stationary rigidity	Stationary L^3 class.	A stationary finite- L^3 self-similar profile must vanish.
Albritton–Barker endpoint ancient Liouville theorem	Uniformly L^3 -tight class.	Tightness plus boundedness yields the sequence- L^3 hypothesis needed to force zero.

Tool	Where it enters	Why it is chosen
Axisymmetric and weighted-vorticity Liouville mechanisms	Structured/decay class.	They exploit special geometric or weighted decay properties instead of requiring a universal ancient Liouville theorem.

The non-residual Type I conclusion is therefore:

```
normalized Type I Seregin profile
|
+-- in small/stationary/tight/structured class --> V = 0
|
+-- but V has retained compact L^3 mass      --> contradiction
```

The only Type I branch left after this point is the residual branch.

15. THE RESIDUAL TYPE I BRANCH: WHY A SECOND PAPER IS NEEDED

The residual paper, `paperIV_residual_branch.tex`, begins after the setup paper has removed the small, stationary L^3 , uniformly tight, and closed structured/decay alternatives. A residual candidate is therefore a bounded normalized centered ancient Seregin profile with retained local concentration which has avoided all direct Liouville classes.

The key idea of the residual paper is that this remaining class is still not a single amorphous object. It has terminal geometry. The proof refines the residual state space by asking ordered questions about symmetry, rotation, stationary hulls, affine/parasitic modes, tail compactness, active loci, diffuse defects, and terminal recurrence.

The residual decomposition is:

```
coarse Type I residual class
|
+-- axisymmetric bounded-circulation residual class C_ax
+-- rotational relative-equilibrium class C_rot
+-- degenerate stationary-hull class C_stat
+-- affine/parasitic quotient stratum C_aff
+-- log-diffuse critical-tail alternative C_logdiff
+-- Young critical-tail alternative C_young
+-- coherent homogeneous critical tail C_homcrit
+-- coherent log-periodic critical tail C_logper
+-- coherent aperiodic critical tail C_apcrit
+-- generic terminal residual C_gen
```

Again, this is an ordered partition. The proof does not need the classes to be intrinsically disjoint in every possible descriptive sense. It needs a deterministic routing rule: after earlier strata have been removed, the first remaining terminal behavior determines the next stratum.

15.1. Residual Type I Taxonomy.

Residual stratum	What it represents	Main closure mechanism
C_{ax} : axisymmetric bounded-circulation residual	The Seregin profile has axisymmetric geometry and bounded centered angular circulation, but was not already removed by the setup structured class.	A centered angular-circulation equation plus axis-absorption Liouville theorem forces the profile into a no-swirl/lower structured class, contradicting residuality.
C_{rot} : rotational relative equilibrium	The profile is stationary only modulo rotation, producing a local Coriolis-type transport structure.	A localized Coriolis-flux identity and local concentration recentering show that either a lower compact profile is produced or the profile falls into an endpoint critical class already closed. No global Coriolis-Leray Liouville theorem is used.

Residual stratum	What it represents	Main closure mechanism
$\mathcal{C}_{\text{stat}}$: degenerate stationary hull with retained concentration	Time-translation hulls of the ancient trajectory contain degenerate stationary-family elements that still carry local concentration.	Scale reselection and the terminal local concentration theorem rule out stationary-family concentration profiles that occur as Seregin descendants after lower classes are removed.
$\mathcal{C}_{\text{aff}}$: affine/parasitic quotient stratum	The residual behavior is an affine or parasitic mode generated by gauge normalization rather than a genuine retained nonlinear residual profile.	Oscillatory-entry normalization removes the quotient stratum: no retained normalized representative remains there.
$\mathcal{C}_{\text{log diff}}$: log-diffuse tail	Critical activity escapes through logarithmic annuli without becoming a compact active profile.	The branch is reduced to hidden-scale compactness and the minimal mesoscopic-scale reduction theorem.
$\mathcal{C}_{\text{Young}}$: Young critical-tail alternative	Critical tail mass survives only as a Young-measure or variance defect rather than a strong local profile.	Frechet–Kolmogorov/Young-measure compactness reduces the defect to the same minimal mesoscopic-scale theorem.
$\mathcal{C}_{\text{homcrit}}$: coherent homogeneous tail	The terminal tail becomes a coherent (-1) -homogeneous critical profile.	Bounded-origin realization and local boundedness contradict the retained residual profile.
$\mathcal{C}_{\text{log per}}$: coherent log-periodic tail	The tail is coherent and periodic in logarithmic scale.	The coherent critical-tail closure excludes realized log-periodic critical tails.
$\mathcal{C}_{\text{apcrit}}$: coherent aperiodic tail	The tail has a compact log-translation hull but no nonzero log period.	The aperiodic hull is closed by the same coherent critical-tail rigidity mechanism.
$\mathcal{C}_{\text{gen}}$: generic terminal residual	Everything left after all lower residual alternatives and tail alternatives have been removed.	Terminal concentration analysis, active path-space recurrence, finite separated-profile exclusion, parasitic-free mildness, sequence- L^3 completeness, and the endpoint ancient Liouville theorem force zero.

15.2. Why the Residual Tools Are Different. The residual branch is difficult because a direct global L^3 bound or global Liouville theorem is not available at the start. The residual paper therefore builds such a conclusion only after eliminating the ways it could fail.

This is a PDE point worth emphasizing. A residual profile may have good local boundedness and retained compact mass while still failing every direct global critical-space hypothesis. The proof does not pretend otherwise. It first removes affine/parasitic modes, diffuse tails, coherent critical tails, and active descendant chains. Only after these exits are closed does the generic residual branch inherit the mildness and sequence- L^3 structure needed for the endpoint ancient Liouville theorem.

The main residual tools are:

Residual tool	Purpose
Ancestor-realization and heredity	Ensures that descendants produced by recentering, blow-down, or terminal extraction are still connected to the original singularity-generated profile. This prevents the proof from proving a theorem about an auxiliary object that is not actually inherited by the original branch.
Terminal local concentration theorem	Classifies where retained local mass can go after lower classes are removed. It excludes pure local vanishing, pure exterior escape, and noncompact terminal alternatives as sole carriers of the retained mass.
Active-locus extraction	Identifies compact regions where a fixed positive amount of critical activity persists. These regions become the nodes of the terminal state-space graph.

Residual tool	Purpose
Diffuse-defect compactness	Handles residual mass that does not localize into finitely many active cores. It distinguishes active regeneration, affine/parasitic collapse, and critical diffuse tails.
Minimal mesoscopic-scale reduction	Prevents hidden critical-tail defects from surviving only between visible scales. It routes log-diffuse and Young alternatives into active, affine, or coherent critical-tail strata.
Critical-tail compactification	Converts noncompact tail behavior into a compact log-scale dynamical problem with homogeneous, log-periodic, and aperiodic alternatives.
Active path-space recurrence	Rules out infinite chains or loops of retained active descendants.
Finite separated-profile exclusion	Prevents retained concentration from splitting into a finite separated family that avoids all lower strata.
Parasitic-free mildness inheritance	After affine/parasitic modes are removed, the physical pullback inherits the mild bounded ancient structure needed for endpoint Liouville theory.
Sequence- L^3 completeness	Converts terminal indecomposability into the sequence- L^3 condition needed by Albritton–Barker.

The final generic residual closure is conceptually:

```

generic residual profile
|
|  --- if active compact concentration remains
|      |  --- finite/infinite descendant analysis
|      |  --- recurrence and separated-family exclusions
|      |
|  --- if diffuse or tail concentration remains
|      |  --- diffuse-defect trichotomy
|      |  --- mesoscopic-scale reduction
|      |  --- critical-tail closure
|      |
|  --- after all lower exits are closed
|      |  --- profile is parasitic-free and sequence- $L^3$ 
|      |  --- Albritton--Barker endpoint Liouville
|      |  --- profile is zero
|      |  --- contradiction with retained local mass

```

This is why the residual proof is efficient. It does not prove a universal Liouville theorem for every bounded centered ancient residual profile. It forces any residual profile that avoids lower strata to become endpoint Liouville-admissible.

16. THE TYPE I ASSEMBLY

Combining `proof_setup.tex` and `paperIV_residual_branch.tex`, the Type I singularity exclusion has the following structure:

Step	Input	Output
Local concentration	$z_* = (x_*, T)$ singular and local pointwise Type I.	A positive local velocity concentration sequence.
No-escape bridge	Endpoint weak-Serrin/local Type I estimates.	The concentration sequence is admissible for Seregin extraction.
Seregin extraction	Admissible Type I bridge sequence.	A smooth bounded centered ancient limit V with retained compact L^3 mass.
Setup class exhaustion	Definition of $\mathcal{S}$.	V is in a non-residual Liouville class or in the residual class.
Non-residual closure	Perturbative, stationary, tight, and structured Liouville mechanisms.	Non-residual classes force $V = 0$, impossible.
Residual closure	Refined residual decomposition in <code>paperIV_residual_branch.tex</code> .	Residual class is empty, so V cannot exist.

Thus a local pointwise Type I singularity cannot occur once the setup and residual papers are combined.

17. THE TYPE II BRANCH: WHAT HAS TO BE CLASSIFIED

The Type II paper starts from the other side of the first dichotomy: positive local critical concentration is present, but no local Type I scale-invariant bound holds. The branch is more geometric than the Type I branch because no single self-similar scale is fixed by the Type I envelope. The concentration may move, split, collapse, or interact across scales.

The Type II paper therefore selects local windows around retained cores and uses repaired gauges. A typical represented branch satisfies a modulated renormalized Navier–Stokes equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

Here $a(\tau)$ records scale modulation and $b(\tau)$ records translation modulation. The repaired gauge chooses the scale and center so that the selected core stays in a controlled coordinate system and the modulation terms are not arbitrary gauge artifacts.

The repaired chart is also designed to preserve the parabolic character of the equation: the time reparametrization is tied to the spatial scale so that the viscosity coefficient remains fixed in the renormalized equation. This is why the Type II analysis can still use local energy, Caccioppoli, pressure reconstruction, and compactness estimates in the transformed variables.

17.1. Why Repaired Gauges Are Used. Without a repaired gauge, a Type II concentration branch can look noncompact for trivial reasons: the core center moves, the selected scale drifts, or two nearby possible cores compete for the normalization. Repaired gauges separate these effects:

Problem	Repaired-gauge response
Moving concentration center	Track a center $x_c(\tau)$ and convert motion into a controlled transport term $b(\tau) \cdot \nabla V$.
Changing concentration scale	Track a scale $\lambda(\tau)$ and convert scale drift into $a(\tau)(V + y \cdot \nabla V)$.
Pressure gauge ambiguity	Reconstruct pressure locally modulo functions of time, matching the local energy inequality.
Loss of compactness by bad coordinates	Use moment or localization constraints to pin down a canonical local chart where compactness questions are meaningful.
Gauge degeneracy	Route failure of the gauge into multibubble, cascade, or scale-rigidity alternatives instead of hiding it inside estimates.

The represented Type II branch is not the conclusion of the proof. It is the coordinate system in which the state-space partition becomes precise.

The nontrivial analytic burden is that the represented branch must remain a suitable local Navier–Stokes object after the transformation. The pressure must be reconstructed modulo functions of time, the local energy inequality must survive on compact windows, and the modulation coefficients must be measurable or locally integrable enough for the distributional identities and compactness arguments. These requirements explain why representation failure is itself part of the Type II partition.

18. TYPE II STATE-SPACE PARTITION

The Type II paper decomposes every positive local Type II concentration sequence into local alternatives and closes each exit. The final assembly distinguishes the geometric alternatives from the local cost/carrier interfaces that close the compact-cost channel:

Type II alternative	Meaning	Closure mechanism
Compact single-core vanishing-dissipation branch	One retained core is selected, compact-window estimates hold, pressure is stable, modulation is bounded, and localized dissipation vanishes.	Local compactness gives a spatially constant or bounded selected-window limit. A zero limit contradicts retained concentration; a nonzero bounded selected-window limit contradicts the Type II regime.
Multibubble, cascade, or gauge-degenerate branch	Positive concentration splits into several active profiles, comparable-scale classes, same-point cascades, separated centers, or nonunique/gauge-degenerate core choices.	Active-profile decoupling, terminal profile decoupling, same-point cascade exclusion, separated-point reduction, and the discharged scale-rigidity package eliminate these branches.
Rough-core / loss of windowed H^1 control	The selected core keeps critical mass but lacks the compact-cylinder gradient control needed for the compact argument.	Local Caccioppoli estimates show finite outer $C + D$ control gives inner H^1 control. Failure of H^1 control is routed back to compactness failure or multibubble/cascade alternatives.
Finite-cost scale-collapse branch	The selected scale genuinely collapses but the scale-collapsing cost or absolute scale cost is finite.	Localized cost identities show finite cost is incompatible with a nonvanishing retained core.
Retained compact cost channel	The branch survives the compact-cost validation tests and carries a validated compact Type II cost-divergence conclusion.	Section 7 verifies the local cost-exclusion data on the retained stratum, proves classwise retained compact coverage, and reduces adjacent exits to the canonical carrier/routing package.
Windowwise carrier-routing alternatives	On the canonical moving cutoff, the only nonautomatic moving components after $J_2, \dots, J_5$ are J_1, J_6, J_7 .	J_1 routes to transition/exterior leakage, J_7 routes to positive scale-work and then scale-rigidity unless an earlier label fires, and J_6 is the assigned negative-drift branch.
Scale-rigid terminal state	Scale-collapse or terminal selection produces an autonomous or scale-rigid reduced state.	Virial-dissipative cases are removed by weighted identities; remaining cases are handled by <code>paper3:prop:scale-rigidity-discharged</code> or reduce to compact single-core behavior.
Terminal critical-profile branch	Terminal active-frame analysis requires critical L^3 profile extraction, nonlinear profile stability, and no hidden terminal mass.	<code>paper6a:thm:critical-ns-profile-decomposition-discharged</code> supplies the profile theorem from cited L^3 profile/stability results plus mass-decoupling, Kato-smallness, and hidden-mass exhaustion lemmas.

This can be rendered as:

```

positive local Type II concentration sequence
|
|  +-- can select one nondegenerate retained core?
|  |
|  |  +-- yes: repaired-gauge representation
|  |  |
|  |  |  +-- compact single-core vanishing-dissipation branch
|  |  |  +-- rough-core / windowed  $H^1$  failure
|  |  |  +-- genuine scale-collapse branch
|  |  |  +-- retained compact-cost / carrier-routing branch
|  |  |  +-- scale-rigid terminal state
|  |  |  +-- terminal critical-profile branch
|  |  |
|  |  +-- no: multibubble / cascade / gauge-degenerate branch
|  |
|  +-- every Type II exit is eliminated

```

18.1. Compact Single-Core Branch. This is the Type II branch closest to a simple blow-up profile. A single core is retained, the repaired-gauge representation is valid, and the local velocity-pressure quantities are uniformly controlled on compact cylinders.

If localized dissipation vanishes, compactness forces the limit to be too rigid. Either the limit is zero, which contradicts the positive CKN mass used to select the core, or the limit is a nonzero bounded selected-window profile, which contradicts the premise that no local Type I scale-invariant bound exists.

This branch is efficient because it turns a Type II question into a local compactness contradiction, not a global profile classification.

18.2. Multibubble and Cascade Branches. The multibubble/cascade analysis handles the ways a Type II branch can avoid being a single compact core.

Subtype	Description	Why it is a separate stratum
Same-point comparable-scale multibubble	Several active profiles occur near the same physical point and at comparable scales.	They must be grouped into a compound active core; below-threshold compounds are perturbative, above-threshold compounds are retained.
Same-point strict scale cascade	Active profiles occur at the same point but at scales separating by large factors.	A nested cascade threatens infinite concentration across scales; cascade length and regularity estimates rule it out.
Separated-point multibubble	Active profiles occur at separated centers.	Local decoupling separates their critical L^3 masses and pressure interactions.
Gauge-degenerate branch	The selected scale-center chart is not uniquely or stably determined.	Degeneracy signals competing cores, cascade structure, or scale-rigid behavior and is routed into the corresponding alternatives.
Terminal active-frame failure	Terminal profiles do not decouple in the required way.	The branch is assigned to terminal profile alternatives and closed by the critical profile decomposition package, terminal admissibility, and the discharged scale-rigidity input.

The proof uses critical small-data stability to remove perturbative profile classes. It uses lower mass thresholds to ensure that every active profile consumes a definite amount of critical mass. The current profile package makes this step explicit: bounded terminal L^3_σ sequences admit an orthogonal profile decomposition, cubic L^3 -mass decoupling, Kato-small heat remainders, nonlinear profile perturbation, and an exhaustion clause saying that any residual terminal frame with nonzero critical

mass generates another profile. This makes infinite or uncontrolled active splitting impossible in the finite critical-mass regimes reached by the terminal profile theorem.

18.3. Rough-Core Branch. The rough-core branch represents a possible selected core with enough critical mass to matter but insufficient regularity for compactness. The local Caccioppoli estimate is the correct tool because it converts outer velocity-pressure control into inner gradient control:

```

finite compact-cylinder C + D control
|
v
inner windowed  $H^1$  control

```

Therefore, if the inner H^1 control fails, the branch cannot be a new singularity type. It means that one of the enclosing compact-cylinder controls has failed, and that failure is already part of the compactness-failure, multibubble, or cascade side of the state-space partition.

18.4. Scale-Collapse Branch. Scale collapse is a genuinely Type II phenomenon: the selected concentration scale can shrink relative to the surrounding terminal window in a way not controlled by a Type I rate. The Type II paper assigns scale-collapse branches to a cost-based partition.

Scale-collapse outcome	Meaning	Treatment
Finite scale-collapsing cost	The signed scale-collapse cost accumulated by the branch is finite.	A nonvanishing localized core forces infinite cost, contradiction.
Finite absolute scale cost	Even the absolute amount of scale drift is finite.	Stronger finite-cost exclusion applies.
Thin-drift alternative	No fixed-length selected window carries the uniform drift needed for the autonomous limit.	Routed outside the compact finite-cost branch and into the ordered alternative structure.
Autonomous-modulation alternative	Thick windows exist but modulation does not converge in the way needed for a clean autonomous limit.	Treated as a modulation/compactness alternative rather than a hidden singularity.
Compactness alternative	Local estimates needed for compact autonomous extraction fail.	Routed to compactness failure or rough-core mechanisms.
Nonzero autonomous reduced limit	A terminal autonomous state survives.	Closed by the appropriate rigidity theorem or virial-dissipative exclusion.

The cost method is efficient because it avoids classifying every possible scale-collapse trajectory. If collapse carries retained mass, the cost must diverge. If it does not diverge, the retained core cannot survive. The latest refinement adds the converse local exclusion interface: once a validated nonnegative localized cost diverges on the canonical windowwise moving schedule, corrected monotonicity rules out the fully summable-error case, so the branch must enter one of the final carrier exits J_1, J_6, J_7 . Those exits are now routed to transition/exterior leakage, negative scale drift, or positive scale-work/scale-rigidity.

For a PDE expert, the useful analogy is that the cost plays the role of a localized monotonicity defect adapted to a moving scale-center frame. It is not introduced as a new global norm. It measures exactly the dissipation and scale drift that would be required for a compact retained core to collapse through renormalized scales, while the corrected-energy argument prevents divergent cost from being treated as a separate final assumption.

18.5. Retained Compact Cost and Carrier Routing. The retained compact-cost layer is the part of the Type II proof that was most refined in the latest pass. The older architecture treated adjacent compact cost exits and cost-divergence exclusion as broad interfaces. The current paper splits them into explicit local tests.

On the retained compact stratum, local cost-exclusion data are verified rather than assumed. Classwise retained compact coverage is discharged by the ordered validation theorem that prevents

unrecorded cost escape. Adjacent compact-cost closure is then reduced to carrier completion, and carrier completion is tested through componentwise fixed and moving finite-error estimates.

On the canonical windowwise moving family, four moving components are automatic: viscous cutoff, convection, pressure, and translation. The remaining components have state-space meanings:

Component	Meaning on the canonical moving family	Route
J_1	Nonintegrable cutoff-transition or shell-leakage component.	Normalized carrier-state extraction labels it as transition/exterior leakage.
J_6	Nonintegrable negative scale-drift contribution.	Assigned to the Section 6 negative-drift/scale-collapse branch.
J_7	Nonintegrable positive scale-shell work.	Routed to positive scale-work, then to scale-rigidity unless an earlier exterior, recentering, or negative-drift label has already fired.

This matters because the proof no longer needs to turn nonintegrability into a pointwise or unit-window lower bound. Diffuse nonintegrable tails are converted into normalized local carrier states, and those states carry the geometric label needed by the ordered partition.

18.6. Terminal Type II States. The Type II paper also records terminal profile and critical-annulus classifications. These distinguish finite critical mass, infinite tail, vanishing critical norm, exterior regularity, and active terminal profiles. The role of these classifications is to prevent terminal escape from becoming a hidden sixth alternative. A branch that tries to avoid the compact single-core mechanism must either regenerate active concentration, enter a scale-rigid state, or lose one of the explicit analytic inputs. It is not left unclassified.

The terminal critical-profile input is also now explicit. The paper proves the critical Navier–Stokes profile package from the standard L^3 profile decomposition and nonlinear profile perturbation theory, plus internal lemmas for L^3 -mass decoupling, heat-flow Kato smallness, and diagonal extraction of hidden terminal mass. Thus “no hidden terminal profile” is not a separate terminal assumption in the current architecture; it is the exhaustion clause of the critical profile theorem.

19. TOOL MAP ACROSS THE WHOLE PROOF

Tool	Used in	What it proves	Why it is efficient
CKN epsilon-regularity	First gate in <code>proof_setup.tex</code> and Type II entry.	Vanishing critical mass implies regularity.	It removes the easiest branch before blow-up analysis begins.
Velocity epsilon-regularity	Local concentration setup.	Velocity concentration must persist at a singular point.	It provides a velocity mass token that survives Seregin extraction.
Endpoint weak-Serrin / local Type I estimate	Type I bridge.	Local pointwise Type I implies no-escape and local energy control.	It converts a pointwise envelope into compactness data.
Seregin extraction	Type I branch.	Produces bounded ancient profiles from Type I singularities.	It turns singularity analysis into ancient-solution classification.
Centered self-similar variables	Type I branch.	Produces the centered autonomous ancient equation.	Time translation becomes the natural compact-dynamical symmetry.

Tool	Used in	What it proves	Why it is efficient
Perturbative Liouville theory	Small Type I class.	Small ancient profiles vanish.	It closes profiles near zero without global classification.
Stationary L^3 rigidity	Stationary Type I class and some terminal states.	Nonzero stationary critical profiles are impossible in the selected stationary strata.	It uses a sharp theorem exactly where stationarity has been selected.
Uniform L^3 -tightness and endpoint Liouville	Tight Type I class and final generic residual closure.	Tight/sequence- L^3 bounded ancient profiles vanish.	It turns tail control into a decisive Liouville contradiction.
Axisymmetric circulation equations	Structured Type I and residual axisymmetric strata.	Bounded-circulation or no-swirl profiles fall to zero/lower classes.	It uses scalar maximum-principle structure unavailable in generic 3D flow.
Weighted-vorticity Lyapunov functionals	Structured Type I class.	Small weighted vorticity states are ancient-rigid.	It closes perturbative coherent vortical structures.
Repaired-gauge representation	Type II branch.	Converts moving scale-center cores into a modulated local PDE.	It distinguishes real noncompactness from coordinate drift.
Local pressure reconstruction	Type I compactness and Type II repaired gauges.	Controls pressure modulo functions of time on compact windows.	It keeps pressure terms compatible with local energy and compactness.
Caccioppoli estimates	Type II rough-core branch.	Outer critical control gives inner H^1 control.	It routes roughness into explicit compactness failure instead of a new branch.
Multibubble/profile decomposition	Type II and residual terminal analysis.	Separates active profiles by scale and center; in the Type II terminal setting, the critical L^3_σ profile package is now discharged by cited decomposition/stability theorems plus internal exhaustion lemmas.	It makes profile interaction countable and thresholded, and prevents hidden terminal mass from remaining after extracted profiles are removed.
Cost and scale-drift identities	Type II scale collapse and compact-cost closure.	Retained cores force infinite scale-collapse cost; corrected monotonicity plus canonical windowwise routing converts divergent localized cost into exclusion or named carrier exits.	It excludes collapse without a full autonomous classification and removes cost-divergence as a final assumption.
Canonical windowwise carrier states	Retained compact Type II cost channel.	Converts nonintegrable moving-error tails into normalized carrier labels for J_1, J_6, J_7 .	It routes diffuse carrier failures without assuming persistent pointwise lower bounds.
Active-locus and diffuse-defect compactness	Residual Type I.	Classifies compact versus diffuse terminal activity.	It prevents residual mass from hiding at infinity or between scales.
Minimal mesoscopic-scale theorem	Residual Type I critical tails.	Routes hidden-scale defects to active, affine, or coherent tail strata.	It closes log-diffuse and Young defects with one common mechanism.

Tool	Used in	What it proves	Why it is efficient
Active path-space recurrence	Generic residual Type I.	Excludes infinite retained descendant chains and recurrent loops.	It prevents endless rerouting of the same residual mass.

20. THE FULL SINGULARITY TAXONOMY

The following table is the architectural classification of singularity candidates across the three papers.

Level	Candidate singularity/profile type	Assigned paper	Status
0	Vanishing CKN concentration	<code>proof_setup.tex</code>	Excluded by epsilon-regularity.
1	Positive concentration with local pointwise Type I envelope	<code>proof_setup.tex</code> plus <code>paperIV_residual_branch.tex</code>	Reduced to Seregin state space, then excluded.
1	Positive concentration with no local Type I envelope	<code>type_II_regularity.tex</code>	Excluded by the closed Type II state-space assembly.
Type I	Small-amplitude Seregin profile	<code>proof_setup.tex</code>	Excluded by perturbative Liouville theorem.
Type I	Stationary L^3 Seregin profile	<code>proof_setup.tex</code>	Excluded by stationary self-similar rigidity.
Type I	Uniformly L^3 -tight Seregin profile	<code>proof_setup.tex</code>	Excluded by endpoint ancient Liouville after physical pullback.
Type I	Liouville-covered structured/decay profile	<code>proof_setup.tex</code>	Excluded by cited structure-specific Liouville theorems.
Type I residual	Axisymmetric bounded-circulation residual	<code>paperIV_residual_branch.tex</code>	Excluded by centered circulation and axis-absorption.
Type I residual	Rotational relative equilibrium	<code>paperIV_residual_branch.tex</code>	Excluded by local Coriolis-flux and terminal concentration decomposition.
Type I residual	Degenerate stationary hull with retained concentration	<code>paperIV_residual_branch.tex</code>	Excluded by scale reselection and terminal local concentration theorem.
Type I residual	Affine/parasitic quotient stratum	<code>paperIV_residual_branch.tex</code>	Removed by oscillatory-entry normalization.
Type I residual	Log-diffuse critical tail	<code>paperIV_residual_branch.tex</code>	Reduced to and excluded by minimal mesoscopic-scale reduction.
Type I residual	Young critical-tail defect	<code>paperIV_residual_branch.tex</code>	Reduced to the same hidden-scale compactness mechanism.

Level	Candidate singularity/profile type	Assigned paper	Status
Type I residual	Coherent homogeneous/log-periodic/aperiodic critical tails	<code>paperIV_residual_branch.tex</code>	Excluded by critical-tail compactification and bounded-origin realization.
Type I residual	Generic terminal residual	<code>paperIV_residual_branch.tex</code>	Excluded by terminal concentration, recurrence, sequence- L^3 , and endpoint Liouville.
Type II	Compact single-core branch	<code>type_II_regularity.tex</code>	Excluded by local compactness and contradiction with Type II condition.
Type II	Multibubble branch	<code>type_II_regularity.tex</code>	Excluded by active profile decoupling, critical profile decomposition, and thresholded mass counting.
Type II	Same-point cascade	<code>type_II_regularity.tex</code>	Excluded by cascade length/control and the discharged scale-rigidity package.
Type II	Separated active profiles	<code>type_II_regularity.tex</code>	Excluded by terminal decoupling and separated-profile reductions.
Type II	Gauge-degenerate branch	<code>type_II_regularity.tex</code>	Routed into multibubble, cascade, or scale-rigid alternatives and excluded there.
Type II	Rough-core branch	<code>type_II_regularity.tex</code>	Reduced by Caccioppoli to compactness failure/multibubble alternatives.
Type II	Finite-cost scale collapse	<code>type_II_regularity.tex</code>	Excluded by localized cost divergence forced by retained mass.
Type II	Retained compact cost channel	<code>type_II_regularity.tex</code>	Closed by verified local cost-exclusion data, classwise retained compact coverage, and adjacent closure reduced to canonical carrier routing.
Type II	Windowwise transition/positive-scale routing	<code>type_II_regularity.tex</code>	Discharged by normalized carrier-state extraction; J_1 routes to transition/exterior leakage and J_7 routes to positive scale-work/scale-rigidity.

Level	Candidate singularity/profile type	Assigned paper	Status
Type II	Divergent compatible localized cost	<code>type_II_regularity.tex</code>	Excluded by corrected monotonicity plus the J_1, J_6, J_7 canonical routing theorem.
Type II	Terminal hidden critical profile mass	<code>type_II_regularity.tex</code>	Excluded by the critical L^3 profile decomposition, nonlinear profile stability, Kato-small remainder, and hidden-mass exhaustion theorem.
Type II	Thin drift, autonomous modulation, compactness alternatives	<code>type_II_regularity.tex</code>	Routed by the ordered scale-collapse stratification.
Type II	Scale-rigid terminal state	<code>type_II_regularity.tex</code>	Excluded by virial identities or <code>paper3:prop:scale-rigidity-discharged</code> .

21. HOW THE STATE-SPACE PARTITION IS ACTUALLY DONE

The partition is not a philosophical label. It is a sequence of tests.

21.1. Step 1: Test for Critical Concentration. At every candidate singular point, inspect the CKN scale-invariant quantities. If they vanish, epsilon-regularity closes the branch. If not, choose a positive concentration sequence.

21.2. Step 2: Test the Local Type I Envelope. On the concentration sequence, decide whether a local pointwise Type I bound holds. If yes, enter the Type I Seregin extraction. If no, enter the Type II local branch analysis.

21.3. Step 3A: Partition the Type I Seregin Collection. For each normalized Seregin ancient profile:

```

Is it small?
  yes --> small class
  no
Is it stationary with  $L^3$  control?
  yes --> stationary class
  no
Is it uniformly  $L^3$ -tight?
  yes --> tight class
  no
Does it satisfy a listed Liouville-covered structure/decay condition?
  yes --> structured/decay class
  no
residual class

```

The order matters only as a routing convention. If a profile satisfies more than one removable property, the earliest applicable closure removes it.

21.4. Step 3B: Refine the Type I Residual. For the residual class:

```

Axisymmetric bounded circulation?
yes --> C_ax
Rotational relative equilibrium?
yes --> C_rot
Degenerate stationary hull with retained concentration?
yes --> C_stat
Affine/parasitic quotient behavior?
yes --> C_aff
Log-diffuse or Young critical-tail defect?
yes --> C_logdiff or C_young
Coherent critical tail?
yes --> C_homcrit / C_logper / C_apcrit
None of the above?
yes --> C_gen

```

Each "yes" branch has a closure theorem. The "none of the above" branch is not ignored. It is the generic residual class, and it is closed last using the terminal concentration theorem, descendant heredity, recurrence exclusion, and endpoint Liouville.

21.5. Step 3C: Partition the Type II Branch. For a positive local Type II concentration sequence:

```

Can a nondegenerate retained core be selected?
no --> multibubble/cascade/gauge-degenerate side
yes
  Build repaired-gauge representation.
  |
  +--- compact single-core with vanishing dissipation?
  |     yes --> compact single-core exclusion
  |
  +--- windowed  $H^1$  control fails?
  |     yes --> rough-core reduction to compactness failure
  |
  +--- genuine scale collapse?
  |     yes --> cost/scale-collapse stratification
  |
  +--- retained compact cost channel?
  |     yes --> verified local cost data + canonical carrier routing
  |
  +--- divergent compatible localized cost?
  |     yes --> corrected monotonicity + J1/J6/J7 routing
  |
  +--- terminal hidden profile mass?
  |     yes --> critical L3 profile decomposition extracts another profile
  |
  +--- scale-rigid terminal state?
  |     yes --> virial or scale-rigidity closure

```

This partition is also ordered. Failure of one representation condition is not a gap; it is routed to an explicit alternative, usually multibubble, cascade, rough-core, compactness failure, or scale-rigidity.

22. WHY THE ARCHITECTURE IS EFFICIENT

The proof is efficient for seven reasons.

First, it uses critical quantities from the beginning. The proof never tracks large non-invariant norms unless they are needed locally to support compactness. The objects C , D , local L^3 mass, and scale-center costs are adapted to Navier–Stokes scaling.

Second, it separates Type I and Type II before doing expensive analysis. Type I has a strong compactness theory leading to bounded ancient solutions. Type II does not. Treating them with the same argument would either lose the Type I structure or over-assume compactness in Type II.

Third, it avoids universal classification theorems. The proof does not require a Liouville theorem for every bounded ancient solution, every rotating profile, or every scale-collapse terminal state.

It proves or cites rigidity only after the state-space partition has produced the exact hypotheses needed.

Fourth, it turns failures into strata. If pressure compactness fails, if a gauge degenerates, if a core becomes rough, if a tail is diffuse, if a critical defect lives between scales, or if a moving carrier component is nonintegrable, the proof does not treat that as a vague technical obstruction. It gives the failure a name and an ordered route in the state space.

Fifth, it preserves ancestry. The residual and Type II arguments repeatedly extract descendants: recentered profiles, terminal profiles, blow-downs, active-locus limits, and hull elements. The proof tracks which descendants are realized by the original singularity-generated branch. This prevents loss of logical force when an auxiliary limit is eliminated.

Sixth, pressure and gauge issues are treated as structural, not cosmetic. In Navier–Stokes compactness arguments the pressure is both nonlocal and gauge ambiguous. The architecture repeatedly normalizes pressure only up to functions of time and insists that local pressure reconstruction be part of the state definition. This is what lets local energy inequalities, CKN quantities, and repaired-gauge limits remain compatible.

Seventh, the proof distinguishes classification from closure. A decomposition theorem says where a branch must go; a closure theorem says that the destination is empty or contradictory. Keeping those roles separate is what makes the dependency graph legible: when a Type II rigidity or compactness input is needed, it appears as the closure of a named stratum, not as an unstated premise hidden inside the decomposition.

23. WHAT IS NOT BEING ASSUMED SILENTLY

Several tempting shortcuts are explicitly avoided by the architecture.

Shortcut not used	Replacement in the proof architecture
A universal Liouville theorem for all bounded ancient Navier–Stokes solutions.	Ordered Type I and residual strata, each closed by a narrower theorem or by terminal reduction to endpoint L^3 .
A global $L_t^\infty L_x^3$ bound for arbitrary residual profiles.	Uniform tightness and sequence- L^3 are produced only after residual tail, affine, and descendant alternatives are removed.
A whole-space Coriolis–Leray Liouville theorem for every bounded rotating profile.	Local Coriolis-flux identities and terminal concentration recentering close the rotational residual stratum.
Treating pressure convergence as automatic.	Pressure gauges, local Calderon–Zygmund reconstruction, and pressure-stable persistence are part of the state definitions.
Treating Type II scale drift as a coordinate nuisance.	Repaired-gauge modulation and scale-cost identities make drift an observable branch of the proof.
Treating compactness failure as a technical remainder.	Compactness failure is stratified into rough-core, multibubble, cascade, terminal, diffuse, and critical-tail alternatives.
Treating nonintegrable moving-cutoff errors as pointwise lower bounds.	Normalized carrier-state extraction routes diffuse J_1 and J_7 tails without requiring persistent unit-window lower bounds.
Assuming cost-divergence exclusion as a final black box.	Corrected monotonicity plus the canonical J_1, J_6, J_7 routing theorem discharges the local cost-divergence criterion.
Assuming terminal profile completeness as an uncited terminal principle.	The Type II paper reduces it to critical L^3 profile decomposition, nonlinear profile stability, Kato-small remainders, and hidden-mass exhaustion.

24. CLOSURE LEDGER

The architecture should be read with the following explicit closure ledger.

Statement	Status in the three-paper architecture
A singular point must carry positive local critical concentration.	Proved from CKN/velocity epsilon-regularity in the setup framework.
Every positive concentration point is Type I or Type II.	Proved as the local blow-up-rate dichotomy.
A local pointwise Type I singularity gives a normalized nonzero bounded centered Seregin ancient profile.	Proved by the setup paper using no-escape and Seregin extraction.
Small, stationary L^3 , uniformly tight, and closed structured/decay Type I classes are impossible.	Proved or reduced to cited Liouville theorems in the setup paper.
The Type I residual class is empty.	Proved by the refined residual decomposition and closure in <code>paperIV_residual_branch.tex</code> .
Type II scale-rigidity closure.	Discharged by <code>paper3:prop:scale-rigidity-discharged</code> after the local compactness and dissipation hypotheses are produced by the Type II routing.
Section 7 retained compact cost coverage and windowwise routing.	Discharged by classwise retained compact coverage, canonical windowwise carrier-state extraction, and the transition/positive-scale routing corollaries.
Local cost-divergence exclusion.	Discharged by corrected monotonicity and the canonical J_1, J_6, J_7 routing theorem.
Critical Navier-Stokes profile decomposition for terminal Type II analysis.	Discharged by cited L^3 profile/stability theorems plus internal mass-decoupling, Kato-smallness, and hidden-mass exhaustion lemmas.
Positive local Type II concentration sequences are excluded.	Proved in <code>type_II_regularity.tex</code> by analytic-data exhaustion, state decomposition, and closure of every local exit.
Full end-to-end regularity conclusion.	Follows from the local concentration dichotomy, Type I setup/residual closure, and closed Type II state-space assembly.

Thus the final combined statement is:

$\wr$ If a finite-energy suitable weak solution had a terminal singular point, local $\wr$ concentration would place it in either the local pointwise Type I branch or $\wr$ the local Type II branch. The Type I branch is eliminated by the Seregin $\wr$ state-space partition in `proof_setup.tex` together with the residual closure $\wr$ in `paperIV_residual_branch.tex`. The Type II branch is eliminated by the $\wr$ closed local state-space assembly in `type_II_regularity.tex`. Therefore no $\wr$ singular point remains.

25. FINAL ASSEMBLY MATRIX

Initial branch	Subdivision	Eliminating paper/tool	Contradiction
No concentration	Vanishing $C + D$ or velocity C .	CKN/velocity epsilon-regularity.	Local boundedness.
Type I	Direct non-residual Seregin classes.	<code>proof_setup.tex</code> : perturbative, NRS, endpoint L^3 , structured Liouville tools.	$V = 0$ contradicts retained L^3 mass.
Type I	Refined residual classes.	<code>paperIV_residual_branch.tex</code> : ordered residual closure.	Residual class is empty, contradicting existence of retained Seregin profile.
Type II	Compact single core.	<code>type_II_regularity.tex</code> : local compactness criterion.	Zero limit loses mass; nonzero bounded limit exits Type II.

Initial branch	Subdivision	Eliminating paper/tool	Contradiction
Type II	Multibubble/cascade/gauge degeneracy.	<code>type_II_regularity.tex</code> : profile decoupling and cascade exclusion.	Active profiles either decouple into excluded cases or force forbidden scale-rigid behavior.
Type II	Rough core.	Caccioppoli estimate.	Roughness is not independent; it routes to compactness failure or multibubble/cascade.
Type II	Scale collapse.	Cost divergence, corrected monotonicity, canonical carrier routing, and autonomous-limit stratification.	Finite cost contradicts retained mass; divergent compatible cost routes to J_1, J_6, J_7 ; terminal scale-rigid states are closed by virial/scale-rigidity inputs.
Type II	Terminal profile incompleteness.	Critical L^3 profile decomposition and hidden-mass exhaustion.	Any hidden terminal critical mass creates another profile, so no unrecorded profile branch remains.

26. PER-NODE CAPSULE TEMPLATES

This section instantiates the proof-capsule template of the strategy paper (*Local Proof Architectures for Critical PDE Regularity*, Appendix “Proof-Capsule Template”) for each node of the decision diagrams above. Every template records the implementing theorem labels in the three source papers, the input contract that the upstream routing must produce, the output contract that the capsule exports, the retained obstruction that survives the capsule, the named failure routes that catch a violated hypothesis, the audit checklist, and the closure status. Templates are grouped to match the ledgers of Section 3 and the diagrams of Section 6, and are intentionally compact: the prose in the preceding sections is the explanation, the templates here are the contract.

Reading convention. Audit slots use a fixed shorthand: S scaling, G pressure/gauge, C compactness topology, R retained obstruction, D dependencies upstream, N nonlinear passage, L locality of the cylinder/window. Closure status is **verified** (the capsule is fully discharged inside the cited paper), **imported** (the capsule’s content is taken from a cited external theorem), or **transport** (the capsule is an upstream interface rather than a terminal contradiction).

26.1. Global and setup capsules.

- Capsule name: Capsule G1: Vanishing CKN concentration $\rightarrow$ regularity.
- Implementation labels:
 - L1. `p0:thm:ckn`.
 - L2. `p0:thm:no-concentration`.
- Source: `proof_setup.tex`.
- Input contract:
 - H1. Suitable weak solution on a backward parabolic cylinder $Q_R(z_*)$.
 - H2. The CKN scale-invariant quantity $C(z_*, r) + D(z_*, r)$ tends to 0 as $r \rightarrow 0$.
 - H3. Pressure normalized by spatial means on each ball.
- Output contract:
 - O1. Local boundedness of u on a smaller cylinder; z_* is regular.
- Retained obstruction:

- I. None retained on this branch; the candidate singular point is removed.
- Failure routes:
- F1. C+D bounded below by a fixed $\eta_0 \rightarrow$ capsule G3 (positive concentration).
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule G2: Vanishing velocity concentration $\rightarrow$ regularity.
- Implementation labels:
- L1. `p0:thm:velocity-ckn`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. Suitable weak solution near z_* .
- H2. Velocity quantity $C(z_*, r) \rightarrow 0$ along the singular scales.
- Output contract:
- O1. Local regularity at z_* .
- Retained obstruction:
- I. None retained on this branch.
- Failure routes:
- F1. C bounded below $\rightarrow$ capsule G3.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule G3: Positive critical concentration token.
- Implementation labels:
- L1. `p0:thm:singular-positive`.
- L2. `p0:cor:velocity-concentration`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. z_* singular and capsules G1, G2 ruled out.
- Output contract:
- O1. There is $\eta_0 \neq 0$ and a sequence $r_n \rightarrow 0$ with $C(z_*, r_n) + D(z_*, r_n) \geq \eta_0$.
- Retained obstruction:
- I. The fixed lower bound η_0 (the retained CKN/velocity mass token).
- Failure routes:
- F1. None inside this capsule; failure here would re-enter G1/G2.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.

- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport; produces the retained token).
- Capsule name: Capsule G4: Type I / Type II dichotomy at z_* .
- Implementation labels:
- L1. `p0:thm:typeI-typeII-dichotomy`.
- L2. `p0:thm:local-weak-serrin-typeI`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. Capsule G3 token is available at z_* .
- Output contract:
- O1. The branch enters either the local pointwise Type I envelope state or the local Type II state (no such envelope).
- Retained obstruction:
- I. The G3 token is preserved on both outgoing branches.
- Failure routes:
- F1. Type I envelope present $\rightarrow$ capsule TII (Seregin extraction route).
- F2. No Type I envelope $\rightarrow$ capsule TII1 (Type II analytic-data entry).
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (routing).
- Capsule name: Capsule G5: Terminal velocity no-escape.
- Implementation labels:
- L1. `p0:thm:auto-velocity-no-escape`.
- L2. `p0:thm:local-weak-serrin-typeI`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. Local pointwise Type I envelope holds near z_* .
- H2. Positive velocity concentration along $r_n \rightarrow 0$.
- Output contract:
- O1. Velocity mass does not escape into the terminal slice; the bridge sequence is admissible for Seregin extraction.
- Retained obstruction:
- I. Positive velocity mass on the bridge sequence.
- Failure routes:
- F1. Loss of velocity mass into $t=T \rightarrow$ reroute to weak-Serrin estimate restoring admissibility, else to capsule G1/G2.
- Audit checklist:

- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A5. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- Closure status:
- verified (transport).

26.2. Direct Type I capsules.

- Capsule name: Capsule TI1: Seregin ancient profile extraction.
- Implementation labels:
- L1. `p1:thm:extraction`.
- L2. `p0:thm:local-weak-serrin-typeI`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. Admissible Type I bridge sequence (capsules G3-G5).
- H2. Local pressure compactness modulo functions of time.
- H3. Centered self-similar variables $(y, \tau) = (x/\sqrt{-t}, -\log(-t))$.
- Output contract:
- O1. A normalized bounded centered ancient profile V on $(-\infty, 0)$ with retained compact L^3 mass on a fixed compact set.
- Retained obstruction:
- I. Compact L^3 nonvanishing of V (the Type I retained mass token).
- Failure routes:
- F1. No-escape failure $\rightarrow$ capsule G5.
- F2. Pressure incompatibility $\rightarrow$ reroute to a pressure/gauge failure capsule inside the setup (no Liouville theorem invoked yet).
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A6. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport into Type I class ledger).
- Capsule name: Capsule TI2: Small-amplitude Type I closure.
- Implementation labels:
- L1. `p1:thm:classical-classes`.
- Source: `proof_setup.tex`.
- Input contract:

- H1. Normalized centered ancient profile V from TI1.
- H2. V is small in the relevant scale-invariant topology.
- Output contract:
- O1. $V = 0$ by perturbative bounded ancient Liouville.
- Retained obstruction:
- I. Compact L^3 mass from TI1.
- Failure routes:
- F1. Smallness fails $\rightarrow$ route to TI3/TI4/TI5/TI6.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal: zero contradicts retained mass).
- Capsule name: Capsule TI3: Stationary L^3 Type I closure.
- Implementation labels:
- L1. `p1:thm:classical-classes`.
- L2. `p2:thm:NRS`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. V (or a time translate) is stationary with critical L^3 control.
- Output contract:
- O1. $V = 0$ by NRS stationary self-similar rigidity.
- Retained obstruction:
- I. Compact L^3 mass.
- Failure routes:
- F1. Stationarity fails $\rightarrow$ route to TI4/TI5/TI6.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule TI4: Uniformly L^3 -tight Type I closure.
- Implementation labels:
- L1. `p1:thm:tight-liouville`.
- L2. `p2:thm:tight-liouville`.
- L3. `p2:thm:AB-endpoint`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. V 's L^3 mass uniformly tight in space along ancient time.
- H2. Physical pullback gives endpoint sequence- L^3 structure.

- Output contract:
- O1. $V = 0$ by Albritton-Barker endpoint ancient Liouville.
- Retained obstruction:
- I. Compact L^3 mass.
- Failure routes:
- F1. Tightness fails $\rightarrow$ route to TI5/TI6.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule TI5: Structured / decay-covered Type I closure.
- Implementation labels:
- L1. `p3:thm:axisymmetric-liouville`.
- L2. `p3:thm:typeI-zero-hypotheses`.
- L3. `p1:thm:known-structure-decay-liouville`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. V exhibits axisymmetry, no-swirl, weighted-vorticity smallness, or one of the listed pointwise scale controls.
- Output contract:
- O1. $V = 0$ by the matching specialized Liouville theorem.
- Retained obstruction:
- I. Compact L^3 mass.
- Failure routes:
- F1. None of the structures hold $\rightarrow$ route to TI6 (residual class).
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule TI6: Direct-class exhaustion $\rightarrow$ residual hand-off.
- Implementation labels:
- L1. `p1:thm:remaining-criterion`.
- Source: `proof_setup.tex`.
- Input contract:
- H1. V fails the small, stationary, tight, and structured/decay tests.
- Output contract:
- O1. V is a normalized residual Seregin profile, handed to paperIV via `hyp:base-seregin-hypotheses`.

- Retained obstruction:
- I. Compact L^3 mass plus residual ancestry.
- Failure routes:
- F1. None: this is a definitional handoff into the Type I residual graph.
- Audit checklist:
- A1. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A2. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport into residual ledger).

26.3. Type I residual capsules.

- Capsule name: Capsule R0: Imported setup package.
- Implementation labels:
- L1. `thm:imported-setup-results`.
- L2. `hyp:base-seregin-hypotheses`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Output of TI6 (residual normalized Seregin profile from setup).
- Output contract:
- O1. The same object is studied throughout paperIV: pressure gauge, retained mass, ancestry, and terminal suitability are fixed.
- Retained obstruction:
- I. Compact L^3 mass plus residual ancestry, transported intact.
- Failure routes:
- F1. Object mismatch with setup $\rightarrow$ reject the residual analysis (does not occur because the import is by reference).
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- imported (interface).
- Capsule name: Capsule R0b: Ancestor-realization and descendant heredity.
- Implementation labels:
- L1. `thm:ancestor-realization-inheritance`.
- L2. `thm:descendant-heredity`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Output of R0.
- H2. Any recentered limit, hull element, tail state, or terminal profile that the proof intends to close.
- Output contract:
- O1. Each such auxiliary object is sequence-realized as a descendant of the original residual branch and inherits the ancestry hypothesis.
- Retained obstruction:

- I. Residual ancestry of every descendant.
- Failure routes:
- F1. Auxiliary object not realized as a descendant $\rightarrow$ the closure theorem for it cannot be applied to the original branch (the proof never invokes it).
- Audit checklist:
- A1. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A2. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- Closure status:
- verified (transport).
- Capsule name: Capsule R1: Axisymmetric bounded-circulation residual.
- Implementation labels:
- L1. `thm:R1-exclusion`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. R0/R0b outputs.
- H2. Lower direct Type I classes already removed.
- H3. The residual profile is axisymmetric with bounded centered circulation.
- Output contract:
- O1. Centered angular-circulation closure plus axis absorption forces the profile into a no-swirl/lower structured class, contradicting residuality.
- Retained obstruction:
- I. Compact L^3 mass and residual ancestry.
- Failure routes:
- F1. Axisymmetry fails $\rightarrow$ route to R2.
- F2. Falls into a lower closed structured class $\rightarrow$ capsule TI5 closes.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal in axisymmetric stratum).
- Capsule name: Capsule R2: Rotational relative equilibrium.
- Implementation labels:
- L1. `thm:terminal-stratification`.
- L2. `thm:R3-no-attainable-degenerate-family` (rotational sub-route).
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Profile stationary modulo rotation; local Coriolis transport structure.
- H2. R1 ruled out.
- Output contract:

- O1. Local Coriolis-flux identity plus terminal concentration recentering either reroute the branch to a closed stratum or force a lower compact profile already excluded.
- Retained obstruction:
 - I. Retained concentration on the rotating frame.
- Failure routes:
 - F1. Rotational structure absent $\rightarrow$ route to R3.
 - F2. Reduction to lower compact profile $\rightarrow$ closed earlier.
- Audit checklist:
 - A1. Scaling: every term in the local identity transforms with its declared critical dimension.
 - A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
 - A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
 - A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
 - A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
 - verified (terminal in rotational stratum).
- Capsule name: Capsule R3: Degenerate stationary hull.
- Implementation labels:
 - L1. `thm:R3-no-attainable-degenerate-family`.
 - L2. `thm:R3S-terminal-theorem`.
 - L3. `cor:R3-closure`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
 - H1. R1, R2 ruled out.
 - H2. A time-translation hull contains a degenerate stationary-family element with retained concentration.
- Output contract:
 - O1. The stationary-hull residual class is empty; in particular no realized active stationary carrier survives the terminal reduction.
- Retained obstruction:
 - I. Retained concentration on the hull element.
- Failure routes:
 - F1. Hull element loses ancestry $\rightarrow$ capsule R0b prevents the closure being misapplied; reroute to lower strata.
- Audit checklist:
 - A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
 - A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
 - A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
 - A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- Closure status:
 - verified (terminal in stationary-hull stratum).
- Capsule name: Capsule R4: Affine / parasitic quotient.
- Implementation labels:

- L1. `thm:affine-normalization-dichotomy`.
- L2. `thm:oscillatory-entry-normalization`.
- L3. `thm:covariant-observer-calculus`.
- L4. `thm:closed-nonaffine-activity`.
- L5. `thm:escaping-covariant-hull-compactness`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. R1-R3 ruled out.
- H2. Apparent residual activity may be generated by an affine/parasitic normalization mode rather than genuine retained nonlinear mass.
- Output contract:
- O1. Oscillatory representatives carry no retained normalized mass; non- oscillatory representatives drop into a lower closed stratum.
- Retained obstruction:
- I. Genuine retained mass after the affine/parasitic quotient is removed.
- Failure routes:
- F1. Quotient does not capture the activity $\rightarrow$ non-affine activity stays closed under the observer calculus and proceeds to the tail/terminal capsules.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal in affine/parasitic stratum).
- Capsule name: Capsule R5a: Recurrent tail-core residual.
- Implementation labels:
- L1. `thm:recurrent-tail-core-rigidity`.
- L2. `thm:terminal-exhaustion-main`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. R1-R4 ruled out.
- H2. Tail activity recurs without entering a compact terminal profile.
- Output contract:
- O1. Either a compact terminal profile is extracted (route to R6) or the recurrence is forbidden by tail-core rigidity.
- Retained obstruction:
- I. Retained mass in the recurrent tail.
- Failure routes:
- F1. Recurrence is genuine and rigid $\rightarrow$ contradiction with retained mass.
- F2. Compact terminal profile present $\rightarrow$ route to R6.
- Audit checklist:
- A1. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.

- A2. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- Closure status:
- verified.
- Capsule name: Capsule R5b: Log-diffuse critical tail.
- Implementation labels:
- L1. `thm:log-diffuse-from-hidden-scale`.
- L2. `thm:critical-tail-compactification-extraction`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. R1-R4 ruled out.
- H2. Critical activity escapes through logarithmic annuli.
- Output contract:
- O1. The log-diffuse defect reduces to hidden-scale compactification and then to a coherent critical-tail state.
- Retained obstruction:
- I. Critical tail mass.
- Failure routes:
- F1. Hidden-scale variance defect realized $\rightarrow$ route to R5c (Young defect).
- F2. Coherent critical tail $\rightarrow$ route to R5d.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule R5c: Young critical-tail / variance defect.
- Implementation labels:
- L1. `thm:first-bad-mesoscopic-reduction`.
- L2. `thm:no-hidden-scale-variance-realized-defects`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Tail mass survives only as a Young/variance defect.
- Output contract:
- O1. Realized hidden-scale variance defects do not exist; mass reduces to the first bad mesoscopic scale and then to coherent critical-tail analysis (R5d).
- Retained obstruction:
- I. Tail Young-measure mass.
- Failure routes:
- F1. Defect realized $\rightarrow$ excluded by L2.
- F2. Defect compactifies coherently $\rightarrow$ R5d.
- Audit checklist:

- A1. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule R5d: Coherent critical-tail rigidity (homogeneous / log-periodic /).
- Implementation labels:
- L1. `thm:coherent-critical-tail-classification`.
- L2. `thm:bounded-origin-realization-coherent-tails`.
- L3. `thm:realized-critical-tail-rigidity`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. R5b/R5c hand off a compactified coherent critical-tail state.
- Output contract:
- O1. The tail is homogeneous, log-periodic, or aperiodic; in every case the realized tail is incompatible with the bounded residual origin.
- Retained obstruction:
- I. Tail critical mass.
- Failure routes:
- F1. Tail is not coherent $\rightarrow$ route back to R5c (variance) or R6 (generic).
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal in critical-tail stratum).
- Capsule name: Capsule R6a: Terminal concentration extraction.
- Implementation labels:
- L1. `thm:maximal-active-family`.
- L2. `thm:terminal-exhaustion-main`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. All lower residual and tail strata closed.
- H2. Residual mass remains.
- Output contract:
- O1. A maximal active family and an exhaustive terminal alternative decomposition (active / nonactive / diffuse / mixed).
- Retained obstruction:
- I. Retained terminal mass on the active family.
- Failure routes:
- F1. Active family empty $\rightarrow$ capsule R6b (inactive exclusion).

- F2. Diffuse/mixed alternative $\rightarrow$ R6c.
- F3. Active profile present $\rightarrow$ R6d.
- Audit checklist:
- A1. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- Closure status:
- verified (transport).
- Capsule name: Capsule R6b: Terminal inactive / nonconcentrating exclusion.
- Implementation labels:
- L1. `thm:terminal-inactive-exclusion`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Terminal extraction (R6a) yields no active compact concentration.
- Output contract:
- O1. Retained mass cannot vanish: this branch contradicts the residual mass token.
- Retained obstruction:
- I. Residual mass token (preserved through R0/R0b).
- Failure routes:
- F1. Some active compactness re-emerges $\rightarrow$ route to R6d.
- Audit checklist:
- A1. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A2. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule R6c: Diffuse and mixed compact-diffuse terminal exclusion.
- Implementation labels:
- L1. `thm:diffuse-defect-compactness`.
- L2. `thm:recurrent-diffuse-defect-measure`.
- L3. `thm:diffuse-defect-trichotomy`.
- L4. `thm:no-mixed-compact-diffuse`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Terminal mass is diffuse, or a mixed compact-diffuse mass coexists.
- Output contract:
- O1. Diffuse trichotomy reroutes active outputs to R6d, affine outputs to R4, and tail outputs to R5d; mixed compact-diffuse is forbidden.
- Retained obstruction:
- I. Residual diffuse mass.
- Failure routes:
- F1. Active output $\rightarrow$ R6d.

- F2. Affine output $\rightarrow$ R4.
- F3. Tail output $\rightarrow$ R5d.
- Audit checklist:
- A1. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- Closure status:
- verified.
- Capsule name: Capsule R6d: Active terminal profile – separated / descendant / recurrent.
- Implementation labels:
- L1. `thm:no-finite-separated-profile-family`.
- L2. `thm:active-path-space-recurrence`.
- L3. `thm:compact-active-descendant`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:
- H1. Active terminal profile present after R6a.
- Output contract:
- O1. Finite separated families are excluded; descendant chains and recurrence are forbidden; a single indecomposable compact active profile remains.
- Retained obstruction:
- I. Retained mass on the single active profile.
- Failure routes:
- F1. Multiple separated profiles $\rightarrow$ excluded by L1.
- F2. Infinite descendant chain $\rightarrow$ excluded by L2/L3.
- Audit checklist:
- A1. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- Closure status:
- verified (transport into R7).
- Capsule name: Capsule R7: Endpoint L^3 closure of the generic residual.
- Implementation labels:
- L1. `thm:mildness-inheritance-main`.
- L2. `thm:AB-main`.
- L3. `thm:R4-final-closure`.
- L4. `thm:paperIV-residual-closure`.
- L5. `cor:setup-residual-hypothesis-proof`.
- Source: `paperIV_residual_branch.tex`.
- Input contract:

- H1. R1-R6d outputs: a single indecomposable parasitic-free compact active profile descends from the residual branch.
- Output contract:
- O1. The profile is sequence- L^3 complete and mild; Albritton-Barker endpoint ancient Liouville forces $V=0$, contradicting retained mass.
- O2. The setup residual hypothesis is discharged.
- Retained obstruction:
- I. Compact L^3 mass.
- Failure routes:
- F1. Mildness fails $\rightarrow$ route back to R4 (parasitic mode missed).
- F2. Sequence- L^3 completeness fails $\rightarrow$ route back to R6c/R6d.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- A6. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- Closure status:
- verified (terminal; closes the residual class).

26.4. Type II capsules.

- Capsule name: Capsule TH1: Type II analytic-data exhaustion (entry).
- Implementation labels:
- L1. `thm:c1-typeII-branch-exhaustion`.
- L2. `cor:c1-ordered-exhaustion`.
- L3. `paper6:thm:paper0`.
- L4. `paper6:thm:physical-typeII-covered-entry`.
- L5. `paper6a:thm:typeII-analytic-data-exhaustive`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Capsule G3 token at z_* ; capsule G4 routes to "no Type I envelope".
- Output contract:
- O1. The branch has the analytic data (concentration, local energy, pressure control, scale-window data) needed for the local Type II assembly, or it is rerouted into a cost/annulus alternative.
- Retained obstruction:
- I. Positive local Type II concentration token.
- Failure routes:
- F1. Analytic data cannot be produced $\rightarrow$ ordered exhaustion routes to a cost or annulus alternative downstream.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.

- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A6. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule TII2: Repaired-gauge representation.
- Implementation labels:
- L1. `paper6:thm:representation`.
- L2. `thm:c2-representation-implication`.
- L3. `thm:c2r-minimal-representation`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Output of TII1.
- H2. A nondegenerate retained core can be selected with controlled scale $\lambda(\tau)$ and center $x_c(\tau)$; modulation coefficients (a,b) are bounded.
- Output contract:
- O1. A represented suitable branch (V,P,a,b) on Q_{R_*} solving the modulated renormalized NS equation with viscosity preserved by the time reparametrization.
- Retained obstruction:
- I. Retained Type II core mass on Q_{r_*} .
- Failure routes:
- F1. Chart nonunique or unstable $\rightarrow$ capsule TII4 (multibubble/cascade) or capsule TII7 (scale-rigid).
- F2. Pressure or local-energy incompatibility $\rightarrow$ capsule TII3.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A6. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule TII3: Pressure / local-energy compatibility in repaired variables.
- Implementation labels:
- L1. `paper2:thm:pressure-decomp`.
- L2. `paper2:thm:A`.
- L3. `paper2:thm:B`.
- L4. `paper2:thm:C`.
- L5. `paper6a:thm:ac-repaired-gauge-consequences`.

- L6. `paper6a:thm:ac-local-caccioppoli`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Output of TII2.
- Output contract:
- O1. Local pressure decomposition modulo functions of time; local energy inequality survives on compact windows; Caccioppoli on inner cylinders.
- Retained obstruction:
- I. Retained core mass; carries gauge bookkeeping forward.
- Failure routes:
- F1. Decomposition fails $\rightarrow$ reroute to TII4 or TII1 (analytic-data alt.).
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule TII4a: Compact single-core vanishing-dissipation.
- Implementation labels:
- L1. `paper1:thm:single-core-criterion`.
- L2. `paper1:lem:zero-dissipation`.
- L3. `paper1:lem:zero-contradiction`.
- L4. `paper1:lem:nonzero-contradiction`.
- L5. `paper6:thm:single-core`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Outputs of TII2/TII3.
- H2. One retained core persists on Q_{r_*} with vanishing localized dissipation; modulation bounded.
- Output contract:
- O1. The selected-window limit is spatially constant; the zero limit contradicts retained core mass; a nonzero limit reroutes to bounded- window behavior outside the strict Type II branch.
- Retained obstruction:
- I. $C_{V_n}(r_*) \geq \eta_0$ along the selected-window subsequence.
- Failure routes:
- F1. Limit nonzero constant $\rightarrow$ exits Type II (bounded window row).
- F2. Multibubble / cascade competing cores $\rightarrow$ TII4b/TII4c.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.

- A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A5. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A6. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A7. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal in single-core row).
- Capsule name: Capsule TII4b: Same-point comparable-scale multibubble.
- Implementation labels:
- L1. `paper3:thm:multi`.
- L2. `paper8:thm:multibubble-frame-reduction`.
- L3. `paper6:thm:multibubble`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Several active profiles near the same point at comparable scales.
- Output contract:
- O1. Active-frame reduction, thresholded mass counting, and terminal decoupling reduce to single-core behavior or to lower closed strata.
- Retained obstruction:
- I. Aggregate retained core mass.
- Failure routes:
- F1. Strict scale separation $\rightarrow$ TII4c (cascade).
- F2. Separated centers $\rightarrow$ TII4d.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A5. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A6. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule TII4c: Same-point strict scale cascade.
- Implementation labels:
- L1. `paper3:thm:same-point-reduction`.
- L2. `paper8:thm:same-point-scale-separation`.
- L3. `paper8:thm:terminal-decoupling-implies-local-decoupling`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Active profiles at the same point on separated scales.
- Output contract:

- O1. Same-point reduction, scale separation, and terminal decoupling close the cascade or reroute irregular cascades to TII5 (rough core) or TII7 (scale-rigid).
- Retained obstruction:
 - I. Cascade mass.
 - Failure routes:
 - F1. Irregular cascade $\rightarrow$ TII5/TII7.
 - Audit checklist:
 - A1. Scaling: every term in the local identity transforms with its declared critical dimension.
 - A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
 - A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
 - A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
 - A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
 - Closure status:
 - verified (terminal).
 - Capsule name: Capsule TII4d: Separated-point multibubble.
 - Implementation labels:
 - L1. `paper3:thm:separated-point-reduction`.
 - L2. `paper8:thm:terminal-decoupling-implies-local-decoupling`.
 - Source: `type_II_regularity.tex`.
- Input contract:
 - H1. Active profiles at separated centers.
- Output contract:
 - O1. Local pressure and mass decoupling separate the centers; each component returns to TII4a (compact core), TII9 (terminal profile), or to small-data closure.
- Retained obstruction:
 - I. Per-center retained mass.
 - Failure routes:
 - F1. Centers merge $\rightarrow$ reroute to TII4b.
 - Audit checklist:
 - A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
 - A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
 - A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
 - A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
 - A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
 - Closure status:
 - verified.
 - Capsule name: Capsule TII4e: Gauge-degenerate branch.
 - Implementation labels:
 - L1. `paper6:thm:state-decomposition` (closure-by-target-exit).
 - L2. `paper6:thm:representation`.
 - Source: `type_II_regularity.tex`.

- Input contract:
- H1. The selected chart is nonunique or unstable because several cores or scales compete.
- Output contract:
- O1. The branch is routed to TII4b/TII4c/TII4d or to TII7 and closed there.
- Retained obstruction:
- I. Retained core mass.
- Failure routes:
- F1. Target exit fails $\rightarrow$ the upstream state-decomposition is reopened; no hidden gauge artifact remains a closed terminal stratum.
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (routing).
- Capsule name: Capsule TII5: Rough core / windowed H^1 failure.
- Implementation labels:
- L1. `paper4:thm:caccioppoli`.
- L2. `paper7:thm>windowed-h1-failure-exclusion`.
- L3. `paper6:thm:rough-core`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Selected core has critical mass but lacks compact-cylinder gradient control.
- Output contract:
- O1. Caccioppoli converts finite outer C+D control to inner H^1 control; if outer control fails, route to compactness failure, multibubble, or cascade exits.
- Retained obstruction:
- I. Retained core mass.
- Failure routes:
- F1. Outer C+D fails $\rightarrow$ TII4b/TII4c.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A5. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A6. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal in rough-core row).
- Capsule name: Capsule TII6a: Finite scale-collapse cost (exclusion).
- Implementation labels:
- L1. `paper5:thm:cost-exclusion`.

- L2. `paper6a:thm:scale-collapse-drift-cost`.
- L3. `paper6:thm:scale-cost`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Selected scale collapses; scale-collapsing cost (signed or absolute) is finite; retained core mass persists.
- Output contract:
- O1. Finite cost is incompatible with a nonvanishing retained core; contradiction.
- Retained obstruction:
- I. Retained core mass; localized scale cost.
- Failure routes:
- F1. Cost diverges $\rightarrow$ capsule TII8 (cost/carrier subtree).
- F2. Core floor fails $\rightarrow$ TII6b.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule TII6b: Collapse core-floor failure.
- Implementation labels:
- L1. `paper5:thm:collapse-core-floor`.
- L2. `paper6a:thm:collapse-core-floor`.
- L3. `paper6a:lem:typeII-concentration-extraction-routes`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. A collapsing branch would fail to retain a nonzero localized core.
- Output contract:
- O1. Vanishing critical L^3 routes to capsule G1/G2 closure or to terminal compactness.
- Retained obstruction:
- I. Retained core mass token.
- Failure routes:
- F1. Core preserved $\rightarrow$ TII6a.
- Audit checklist:
- A1. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.

- Capsule name: Capsule TII6c: Thin-drift scale-collapse alternative.
- Implementation labels:
 - L1. `paper5:thm:scale-collapse-first-failure-closure`.
 - L2. `paper5:thm:scale-collapse-alternatives`.
- Source: `type_II_regularity.tex`.
- Input contract:
 - H1. No fixed-length selected window carries enough drift for autonomous extraction.
- Output contract:
 - O1. Branch is routed to compactness, rough-core, or carrier exits.
- Retained obstruction:
 - I. Retained core mass on thin-drift schedule.
- Failure routes:
 - F1. Thick window present $\rightarrow$ capsule TII6d (autonomous modulation).
- Audit checklist:
 - A1. Scaling: every term in the local identity transforms with its declared critical dimension.
 - A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
 - A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
 - A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
 - A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
 - verified (transport).
- Capsule name: Capsule TII6d: Autonomous-modulation alternative.
- Implementation labels:
 - L1. `paper5:thm:autonomous-modulation-selection`.
 - L2. `paper5:thm:autonomous-limit`.
 - L3. `paper5:lem:autonomous-scale-collapse-is-scale-rigid`.
- Source: `type_II_regularity.tex`.
- Input contract:
 - H1. Thick windows exist but modulation fails to converge cleanly.
- Output contract:
 - O1. Autonomous reduced-limit extraction succeeds, or compactness fails and reroutes to TII5.
- Retained obstruction:
 - I. Retained mass on autonomous frame.
- Failure routes:
 - F1. Reduced limit nonzero $\rightarrow$ TII7 (scale-rigid).
 - F2. Compactness fails $\rightarrow$ TII5.
- Audit checklist:
 - A1. Scaling: every term in the local identity transforms with its declared critical dimension.
 - A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
 - A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
 - A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.

- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII7: Scale-rigid stratum closure.
- Implementation labels:
- L1. `paper3:prop:scale-rigidity-discharged`.
- L2. `paper3:thm:scale-rigid-single-core-compatibility`.
- L3. `paper3:thm:scale-rigid-stratum-decomposition`.
- L4. `paper3:prop:local-scale-rigidity-exclusion`.
- L5. `paper6:thm:scale-stratification`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. A nonzero autonomous or scale-rigid terminal state survives from TII6c/TII6d.
- Output contract:
- O1. Scale-rigid sub-strata reduce to virial, compact-core, or multibubble closures, all of which are already closed.
- Retained obstruction:
- I. Scale-rigid retained mass.
- Failure routes:
- F1. Unidentified subtype $\rightarrow$ stratum decomposition forces a finite enumeration; no open terminal sub-stratum remains.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal).
- Capsule name: Capsule TII8a: Critical tightness / compact cost validation.
- Implementation labels:
- L1. `paper7:thm:critical-tightness-criteria`.
- L2. `paper7:thm:carrier-validation-retained-stratum`.
- L3. `paper7:thm:classwise-retained-compact-coverage`.
- L4. `paper7:thm:local-cost-to-exclusion`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Branch retains compact tests, local mass floor, and localized cost comparison after TII3.
- Output contract:
- O1. Branch enters the retained compact-cost stratum and a compact cost- divergence conclusion is produced; classwise coverage is complete.
- Retained obstruction:
- I. Retained compact-cost token.
- Failure routes:
- F1. Tightness fails $\rightarrow$ tail/exterior routing (capsule TII11).

- F2. Coverage fails $\rightarrow$ TII8b adjacent closure.
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule TII8b: Adjacent compact-cost / carrier completion.
- Implementation labels:
- L1. `paper7:thm:adjacent-closure-reduced-to-carrier-completion.`
- L2. `paper7:thm:carrier-completion-reduced-to-finite-error.`
- L3. `paper7:thm:fixed-carrier-closure.`
- L4. `paper7:thm:moving-carrier-closure.`
- L5. `paper7:thm:finite-error-selection-reduced-to-annular.`
- L6. `paper7:thm:carrier-package-closes-precARRIER.`
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Output of TII8a; fixed/moving cutoff carrier errors must be decomposed into finite-error identities.
- Output contract:
- O1. Branch reduces to the canonical moving family $J_1, \dots, J_7$; only J_1, J_6, J_7 may carry nonintegrable mass.
- Retained obstruction:
- I. Retained compact-cost token.
- Failure routes:
- F1. Unrecorded leakage $\rightarrow$ excluded by L3/L4.
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule TII8c1: J_1 transition / exterior carrier.
- Implementation labels:
- L1. `paper7:thm>windowwise-transition-routing-certificate.`
- L2. `paper7:thm>windowwise-routing-gives-stratified-routing.`

- L3. `paper7:cor>windowwise-transition-routing-holds`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. The cutoff-transition or shell-leakage component is nonintegrable.
- Output contract:
- O1. Normalized carrier-state extraction routes the branch to exterior regularity (TII11) or transition closure.
- Retained obstruction:
- I. Carrier-state mass on the transition shell.
- Failure routes:
- F1. Exterior regularity applies $\rightarrow$ TII11.
- F2. Transition closure fires $\rightarrow$ closed inside `paper7`.
- Audit checklist:
- A1. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII8c6: J_6 negative scale-drift carrier.
- Implementation labels:
- L1. `paper6a:thm:moving-cutoff-scale-collapse-alternative`.
- L2. `paper6a:thm:finite-infinite-scale-negative`.
- L3. `paper6a:thm:thin-negative-drift-closed`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Negative scale-drift component is nonintegrable.
- Output contract:
- O1. Branch routes to negative-drift scale-collapse and is closed there.
- Retained obstruction:
- I. Carrier-state mass on negative drift.
- Failure routes:
- F1. Negative drift compatible with finite cost $\rightarrow$ reroute to TII6a.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII8c7: J_7 positive scale-shell work carrier.
- Implementation labels:
- L1. `paper7:thm:positive-scale-certificate-enters-scale-rigid`.

- L2. `paper7:thm>windowwise-positive-scale-routing-certificate`.
- L3. `paper7:cor>windowwise-positive-scale-routing-holds`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Positive scale-shell work component is nonintegrable; no earlier exterior, recentering, or negative-drift label fires.
- Output contract:
- O1. Branch routes to scale-rigidity (TII7).
- Retained obstruction:
- I. Carrier-state mass on positive scale-work.
- Failure routes:
- F1. Earlier label fires $\rightarrow$ TII8c1 / TII8c6.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII8d: Cost-divergence exclusion (assembly).
- Implementation labels:
- L1. `paper6a:thm:cost-divergence-routing-dichotomy`.
- L2. `paper6a:thm:cost-divergence-exclusion-discharged`.
- L3. `paper6a:lem:corrected-monotonicity-versus-divergent-cost`.
- L4. `paper6a:thm:fixed-corrected-monotonicity`.
- L5. `paper6a:thm:moving-corrected-monotonicity`.
- L6. `paper6a:thm:fixed-cost-criterion`.
- L7. `paper6a:thm:compact-cost-criterion`.
- L8. `paper6a:thm:moving-replacement`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Retained compact cost forces divergence on the canonical windowwise schedule; corrected monotonicity holds with finite initial corrected energy.
- Output contract:
- O1. Divergent compatible cost cannot remain a free terminal branch; the capsule routes to TII8c1/TII8c6/TII8c7 and closes.
- Retained obstruction:
- I. Carrier cost token.
- Failure routes:
- F1. Some component summable $\rightarrow$ TII8b reroutes; nonintegrable component determines TII8c*.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.

- A4. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A5. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A6. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A7. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal at the cost layer).
- Capsule name: Capsule TII9a: Vanishing terminal critical L^3 .
- Implementation labels:
- L1. `paper6a:lem:typeII-concentration-extraction-routes`.
- L2. `paper6a:lem:terminal-local-critical-mass-routing`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Terminal branch loses critical L^3 mass.
- Output contract:
- O1. Branch routes to nontrivial-critical-core or vanishing-critical exclusion; cannot remain a Type II singular branch.
- Retained obstruction:
- I. Terminal critical mass token.
- Failure routes:
- F1. Critical mass survives $\rightarrow$ TII9b/TII9c.
- Audit checklist:
- A1. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A2. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII9b: Terminal adjacent label.
- Implementation labels:
- L1. `paper6a:thm:terminal-adjacent-label-routing`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Terminal compactness produces a state adjacent to a carrier or scale branch.
- Output contract:
- O1. Routed to the already-closed adjacent exit (TII7 / TII8*).
- Retained obstruction:
- I. Adjacent retained mass.
- Failure routes:
- F1. No adjacent label applies $\rightarrow$ TII9c (residual mass).
- Audit checklist:
- A1. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.

- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII9c: Residual terminal mass / hidden critical profile.
- Implementation labels:
- L1. `paper6a:thm:residual-mass-admissible-state`.
- L2. `paper6a:thm:critical-ns-profile-decomposition-discharged`.
- L3. `paper6a:thm:kato-small-data-critical-stability`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. A terminal branch retains mass outside the recorded compact/core states; terminal frames sampled from the represented orbit are bounded in the critical L^3_σ framework.
- Output contract:
- O1. Critical L^3 profile decomposition extracts every nonzero terminal profile; Kato-small remainders cannot carry hidden mass; no hidden profile branch remains.
- Retained obstruction:
- I. Terminal critical L^3 mass.
- Failure routes:
- F1. Profile extracted but not closed $\rightarrow$ rerun TII9b.
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Nonlinear passage: each nonlinear term passes to the limit in the declared topology, with no implicit Aubin–Lions or whole-space estimate.
- A6. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII10: Retained compact-test closure / terminal local compactness.
- Implementation labels:
- L1. `paper6a:thm:retained-compact-tests-closed`.
- L2. `paper6a:thm:terminal-critical-local-compactness`.
- L3. `paper6a:thm:bounded-terminal-sequences-from-l3`.
- L4. `paper6a:thm:terminal-boundedness-alternatives`.
- L5. `paper6a:thm:terminal-coordinate-frame-construction-profile-selection`.
- L6. `paper6a:thm:terminal-coordinate-frames-sampled`.
- L7. `paper6a:thm:representation-gives-terminal-orbit-sampling`.
- L8. `paper6a:thm:expanded-terminal-typeII`.
- L9. `paper6a:thm:expanded-terminal-data-branchwise`.
- Source: `type_II_regularity.tex`.
- Input contract:

- H1. Retained compact tests survive but are not yet assigned to an existing Type II state; terminal frames are bounded, unbounded, or sampled in incompatible ways.
- Output contract:
- O1. Bounded terminal sequences are constructed; coordinate-frame and orbit-sampling alternatives are exhausted; the branch is assigned to a closed Type II state.
- Retained obstruction:
- I. Retained compact-test mass.
- Failure routes:
- F1. Test cannot be assigned $\rightarrow$ reroute to TII9c.
- Audit checklist:
- A1. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A5. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII11: Exterior terminal concentration.
- Implementation labels:
- L1. `paper8:thm:exterior-stratification`.
- L2. `paper8:thm:exterior-regularity-no-concentration`.
- L3. `paper8:thm:exterior-regular-removal`.
- L4. `paper8:thm:exterior-concentration-routed`.
- Source: `type_II.regularity.tex`.
- Input contract:
- H1. Terminal mass escapes to the exterior region rather than a compact core.
- Output contract:
- O1. Exterior stratification + exterior regularity remove the concentration; routed exterior concentration cannot remain.
- Retained obstruction:
- I. Exterior retained mass.
- Failure routes:
- F1. Concentration recompactifies inside $\rightarrow$ reroute to TII4a.
- Audit checklist:
- A1. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A2. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A3. Compactness topology: the convergence used is strong enough for every nonlinear passage in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified.
- Capsule name: Capsule TII12: Final Type II assembly.

- Implementation labels:
- L1. `paper6:thm:state-decomposition`.
- L2. `paper6:prop:state-elimination`.
- L3. `paper6a:thm:abstract-final-typeII`.
- L4. `paper6a:thm:expanded-terminal-typeII`.
- L5. `paper6:thm:unconditional-no-typeII`.
- L6. `paper6:thm:unconditional-no-typeII-discharged`.
- L7. `paper6:thm:local-typeII-exclusion`.
- L8. `paper6:thm:autonomous-modulation-failure-closed`.
- L9. `paper6:thm:noncanonical-cost-routing-closed`.
- L10. `paper6:thm:active-core-loss-closed`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Every individual Type II exit closed by capsules TII1-TII11.
- Output contract:
- O1. No local Type II singular branch remains.
- Retained obstruction:
- I. None survives the final assembly.
- Failure routes:
- F1. State decomposition reopens $\rightarrow$ the offending capsule is identified and re-audited; fault containment ensures other branches stay closed.
- Audit checklist:
- A1. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A2. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal; closes the Type II layer).

26.5. Cost / carrier subtree micro-capsules. The capsules below refine the cost/carrier subtree at the level of the diagram in Section 6; they are sub-capsules of TII8a-d above and are listed for diagram-level traceability.

- Capsule name: Capsule K1: Corrected-monotonicity check on the validated compact cost.
- Implementation labels:
- L1. `paper6a:thm:fixed-corrected-monotonicity`.
- L2. `paper6a:thm:moving-corrected-monotonicity`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. A validated compact Type II cost has been selected on the retained branch.
- H2. The localized energy identity and finite initial corrected energy are available on the fixed cutoff or on the canonical moving cutoff.
- Output contract:
- O1. A corrected monotonicity inequality holds on the chosen cutoff package.
- O2. The corrected energy controls the compatible localized cost on that package.
- Retained obstruction:
- I. The retained compact-cost token before carrier-component routing.
- Failure routes:
- F1. Fixed-cutoff data with finite fixed error tail route to the fixed monotonicity subcase.
- F2. Canonical moving-cutoff data with summable moving-annulus errors route to the moving monotonicity subcase.

- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Pressure/gauge: the pressure is normalized only modulo functions of time or spatial means, compatibly with all limits.
- A3. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule K2: Finite-vs-divergent total cost dichotomy.
- Implementation labels:
- L1. `paper6a:lem:corrected-monotonicity-versus-divergent-cost`.
- L2. `paper6a:thm:cost-divergence-routing-dichotomy`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. Corrected monotonicity from K1 holds on the canonical windowwise schedule.
- H2. The retained compact cost diverges on every later tail of the active branch.
- Output contract:
- O1. Fully summable moving errors would force finite total compatible cost, contradicting the divergent-cost branch.
- O2. Hence at least one moving-error component is nonintegrable on every later tail, and the branch routes to K3.
- Retained obstruction:
- I. The retained compact-cost token on the canonical moving schedule.
- Failure routes:
- F1. If the corrected-monotonicity hypotheses are not available, reroute to K1 rather than invoking the dichotomy.
- F2. If cost divergence has not yet been validated on the active branch, remain in the upstream compact-cost validation layer rather than entering K2.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (transport).
- Capsule name: Capsule K3: Canonical component selection (J_1, J_6, J_7) .
- Implementation labels:
- L1. `paper7:cor>windowwise-only-transition-and-scale-remain`.
- L2. `paper6a:thm:cost-divergence-routing-dichotomy`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. K2 has produced a nonintegrable moving-error component on every later tail.
- H2. The canonical windowwise moving family has already made $J_2, \dots, J_5$ summable.
- Output contract:
- O1. The failing component is forced into the canonical set $\{J_1, J_6, J_7\}$.

- O2. The branch routes to the matching carrier capsule TII8c1, TII8c6, or TII8c7.
- Retained obstruction:
- I. The retained carrier-component mass on the canonical moving schedule.
- Failure routes:
- F1. Transition/leakage component J_1 fails $\rightarrow$ TII8c1.
- F2. Negative drift or positive scale-work fails $\rightarrow$ TII8c6 / TII8c7.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (routing).
- Capsule name: Capsule K_{final} : Cost-divergence exclusion.
- Implementation labels:
- L1. `paper6a:thm:cost-divergence-exclusion-discharged`.
- L2. `paper6a:cor:cost-divergence-exclusion-unconditional`.
- Source: `type_II_regularity.tex`.
- Input contract:
- H1. The named carrier exits TII8c1, TII8c6, and TII8c7 are closed by their assigned arguments.
- H2. The canonical windowwise schedule and adjacent-stratum closure package are already in force.
- Output contract:
- O1. Cost-divergence exclusion is discharged, so no residual cost-divergence branch remains open.
- Retained obstruction:
- I. The carrier cost token that persists until all named exits are closed.
- Failure routes:
- F1. If a named carrier exit is not yet closed, reroute to the corresponding TII8c* capsule instead of invoking K_{final} .
- F2. If adjacent-stratum closure is unavailable, remain in the ambient Type II closure package rather than claiming final cost-divergence exclusion.
- Audit checklist:
- A1. Scaling: every term in the local identity transforms with its declared critical dimension.
- A2. Locality: the capsule uses only the declared cylinder/window/state, with no hidden whole-space data.
- A3. Retention: the retained obstruction declared in I is preserved through every operation in the capsule.
- A4. Dependencies: every hypothesis is produced by an upstream capsule, with no circular use of a downstream closure.
- Closure status:
- verified (terminal at the carrier subtree).

26.6. Coverage and cross-checks.

- Every node of the four state-space decision trees in Section 6 corresponds to one or more capsules above: global gates (G1–G5), Type I extraction and direct classes (TI1–TI6), Type

I residual strata (R0–R7), Type II representation/closure (TII1–TII12), and the cost/carrier subtree (K1–K_final). Decision diamonds in the diagrams are routing capsules whose output contract names the downstream capsules; box nodes correspond to terminal contradictions or to transport capsules feeding the next state.

- Every implementation label cited above has been verified to occur in one of the three source files (`proof_setup.tex`, `paperIV_residual_branch.tex`, `type_II_regularity.tex`) at the time of writing. One obsolete terminal nonactive-exclusion placeholder from earlier diagram drafts has been corrected to the actual label `thm:terminal-inactive-exclusion`; this correction does not change the proof strategy and does not introduce new assumptions.
- No capsule above introduces a new analytic assumption beyond what is already stated and discharged in the cited theorems. The capsules are a contract-level rewriting of the existing ledger and diagram entries; their purpose is to make the input/output interfaces explicit and locally auditable.

27. COMPLETE THEOREM LABEL REGISTER

This register is regenerated against the current `theorem` environments in the three TeX sources. It is grouped by file so the labels can be cross-checked against the decision trees and the source. Some theorem environments carry non-`thm:` prefixes because the source uses hypothesis-style or lemma-style labels for theorem-environment statements.

27.1. `proof_setup.tex`.

Label

`p0:thm:ckn`
`p0:thm:velocity-ckn`
`p0:thm:no-concentration`
`p0:thm:singular-positive`
`p0:thm:auto-velocity-no-escape`
`p0:thm:typeI-typeII-dichotomy`
`p0:thm:local-weak-serrin-typeI`
`p1:thm:remaining-criterion`
`p1:thm:classical-classes`
`p1:thm:tight-liouville`
`p1:thm:known-structure-decay-liouville`
`p2:thm:tight-compactness-intro`
`p2:thm:minimal-tight-intro`
`p2:thm:tight-liouville-intro`
`p2:thm:stationary-limits-tight`
`p2:thm:NRS`
`p2:thm:minimal-existence`
`p2:thm:compact-hull`
`p2:thm:KB`
`p2:thm:statistical-identity`
`p2:thm:barycenter-covariance`
`p2:thm:AB-endpoint`
`p2:thm:tight-liouville`
`p2:thm:no-tight-normalized-collection`
`p3:thm:axisymmetric-liouville`
`p3:thm:weighted-vorticity-vanishing-intro`
`p3:thm:typeI-consequences`
`p3:thm:knss-noswirl`
`p3:thm:knss-pointwise`
`p3:thm:lzz`
`p3:thm:lrz`
`p3:thm:gallay`

Label

p3:thm:typeI-zero-hypotheses
p3:thm:stated-hypotheses-zero
p1:thm:final-assembly

27.2. paperIV_residual_branch.tex.

Label

thm:imported-setup-results
hyp:base-seregin-hypotheses
thm:ancestor-realization-inheritance
thm:refined-residual-closure
thm:AB-weak-serrin-typeI-estimate
thm:local-typeI-terminal-energy-estimates
thm:local-typeI-entry-or-exclusion
thm:R1-exclusion
thm:covariant-observer-calculus
thm:closed-nonaffine-activity
thm:affine-normalization-dichotomy
thm:escaping-covariant-hull-compactness
thm:recurrent-tail-core-rigidity
thm:maximal-active-family
lem:compact-window-expanding-region-protocol
thm:terminal-exhaustion-main
thm:terminal-inactive-exclusion
thm:diffuse-defect-compactness
thm:diffuse-defect-compactification-construction
thm:recurrent-diffuse-defect-measure
thm:diffuse-defect-trichotomy
thm:critical-tail-compactification-compactness
thm:log-diffuse-from-hidden-scale
thm:first-bad-mesoscopic-reduction
thm:no-hidden-scale-variance-realized-defects
thm:critical-tail-compactification-extraction
thm:coherent-critical-tail-classification
thm:realized-critical-tail-rigidity
thm:oscillatory-entry-normalization
thm:remaining-rigidity-exact-form
thm:bounded-origin-realization-coherent-tails
thm:no-mixed-compact-diffuse
thm:mildness-inheritance-main
thm:AB-main
thm:descendant-heredity
thm:active-path-space-recurrence
thm:compact-active-descendant
thm:no-finite-separated-profile-family
thm:terminal-stratification
thm:generic-terminal-exhaustion
thm:R4-final-closure
thm:paperIV-residual-closure

27.3. type_II_regularity.tex.

Label

paper1:thm:paper0-dichotomy
paper1:thm:single-core-criterion

Label

paper2:thm:ren-NS
 paper2:thm:repaired-existence
 paper2:thm:pressure-decomp
 paper2:thm:mod-sys
 paper2:thm:C
 paper3:thm:minimal-bad-extraction
 paper3:thm:separated-point-reduction
 paper3:thm:scale-rigid-single-core-compatibility
 paper3:thm:scale-rigid-stratum-decomposition
 paper3:thm:multi
 paper4:thm:caccioppoli
 paper5:thm:collapse-core-floor
 paper5:thm:cost-exclusion
 paper5:thm:autonomous-limit
 paper5:thm:scale-collapse-stratification
 paper5:thm:selected-compactness-failure-routed
 paper5:thm:scale-collapse-first-failure-closure
 paper5:thm:compact-scale-collapse-rigidity-discharged
 paper5:thm:scale-collapse-alternatives
 paper7:thm:critical-tightness-criteria
 paper7:thm>windowed-h1-failure-exclusion
 paper7:thm:local-cost-to-exclusion
 paper7:thm:carrier-validation-retained-stratum
 paper7:thm:classwise-retained-compact-coverage
 paper7:thm:pointwise-cost-exclusion
 paper7:thm:noncarrier-exits-route
 paper7:thm:precarrier-reduces-to-carrier
 paper7:thm:fixed-carrier-closure
 paper7:thm:moving-carrier-closure
 paper7:thm:finite-error-selection-reduced-to-annular
 paper7:thm:carrier-completion-reduced-to-finite-error
 paper7:thm:carrier-package-closes-precarrier
 paper7:thm:adjacent-closure-reduced-to-carrier-completion
 paper7:thm>windowwise-transition-routing-certificate
 paper7:thm:positive-scale-certificate-enters-scale-rigid
 paper7:thm>windowwise-positive-scale-routing-certificate
 paper7:thm>windowwise-routing-gives-stratified-routing
 paper7:thm:ambient-adjacent-closure-branchwise
 paper8:thm:same-point-scale-separation
 paper8:thm:multibubble-frame-reduction
 paper8:thm:terminal-decoupling-implies-local-decoupling
 paper8:thm:terminal-windowed-profile-completeness
 paper8:thm:terminal-nonlinear-decoupling
 paper8:thm:small-data-critical
 paper8:thm:active-profile-mass-floor
 paper8:thm:terminal-ckn-regularity
 paper8:thm:exterior-stratification
 paper8:thm:exterior-regularity-no-concentration
 paper8:thm:exterior-regular-removal
 paper8:thm:exterior-concentration-routed
 paper6a:thm:fixed-corrected-monotonicity
 paper6a:thm:fixed-cost-criterion
 paper6a:thm:compact-cost-criterion
 paper6a:thm:moving-corrected-monotonicity
 paper6a:thm:moving-replacement
 paper6a:thm:finite-infinite-scale-negative
 paper6a:thm:collapse-core-floor

Label

```

paper6a:thm:scale-collapse-drift-cost
paper6a:thm:moving-cutoff-scale-collapse-alternative
paper6a:thm:cost-divergence-routing-dichotomy
paper6a:thm:cost-divergence-exclusion-discharged
paper6a:thm:ac-repaired-gauge-consequences
paper6a:thm:ac-local-caccioppoli
paper6a:thm:terminal-adjacent-label-routing
paper6a:thm:residual-mass-admissible-state
paper6a:thm:critical-ns-profile-decomposition-discharged
paper6a:thm:kato-small-data-critical-stability
paper6a:thm:retained-compact-tests-closed
paper6a:thm:thin-negative-drift-closed
paper6a:thm:typeII-analytic-data-exhaustive
paper6a:thm:terminal-critical-local-compactness
paper6a:thm:bounded-terminal-sequences-from-l3
paper6a:thm:terminal-boundedness-alternatives
paper6a:thm:terminal-coordinate-frame-construction-profile-selection
paper6a:thm:terminal-coordinate-frames-sampled
paper6a:thm:representation-gives-terminal-orbit-sampling
paper6a:thm:abstract-final-typeII
paper6a:thm:expanded-terminal-typeII
paper6a:thm:expanded-terminal-data-branchwise
paper6:thm:paper0
paper6:thm:representation
paper6:thm:single-core
paper6:thm:multibubble
paper6:thm:rough-core
paper6:thm:scale-cost
paper6:thm:scale-stratification
paper6:thm:state-decomposition
paper6:thm:physical-typeII-covered-entry
paper6:thm:autonomous-modulation-failure-closed
paper6:thm:noncanonical-cost-routing-closed
paper6:thm:active-core-loss-closed
paper6:thm:unconditional-no-typeII
paper6:thm:unconditional-no-typeII-discharged
paper6:thm:local-typeII-exclusion
thm:c1-typeII-branch-exhaustion
thm:c2-representation-implication
thm:c2r-minimal-representation
thm:c3-local-retained-mass-cost-divergence
thm:c4-cost-comparison

```

The proof architecture is therefore a finite decision tree with terminal contradictions. Its strength is that every possible singularity type is routed before it is eliminated. Its discipline is that every route specifies the analytic input it uses.